
CauSciBench: Can LLMs Automate Causal Inference in Real-World Scientific Research?

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🔗: <https://github.com/causalNLP/CauSciBench>.

Abstract

Identifying and estimating causal relationships from data is a crucial component of empirical research. While large language model-powered tools have shown potential for assisting research workflows, their ability to perform end-to-end causal inference remains underexplored. We introduce **CauSciBench**, a benchmark that puts LLM-powered tools to the test on causality-driven research questions. Unlike previous related benchmarks that focus on coding alone, CauSciBench enables evaluation across the full pipeline of causal inference: from method and variable selection to computation of causal effects and statistical interpretation in the context of real-world research problems. We evaluated 7 frontier models on over 300 queries derived from scientific publications, textbook problems, seminal datasets, and synthetic scenarios. Results show that models consistently perform worse on real datasets, with the key bottleneck being the selection of an appropriate causal inference method.

1. Introduction

Many scientific investigations involve estimating causal relationships between variables of interest (Gangl, 2010;

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Kleinberg & Hripacsak, 2011; Glass et al., 2013; Athey & Imbens, 2017; Imbens, 2024). While practitioners often have access to datasets that can be used to address these questions, the method used to analyze the data and draw causal conclusions depends heavily on the specific context. Unlike machine learning tasks with a well-defined loss function, causal inference does not involve a metric that can be directly optimized to determine the best approach. Instead, the appropriateness of a method depends on the study design, the data collection process, and the assumptions that can be justified in a given setting (Pearl, 2009; Imbens & Rubin, 2015; Peters et al., 2017; Cunningham, 2021; Hernan & Robins, 2025). Together, these factors make causal inference a challenging task, one that requires both strong domain understanding and technical knowledge.

As a motivating example, education policy researchers may be interested in the question: **How much does after-school tutoring improve academic performance?** The answer depends on the nature of the dataset. The most straightforward case is a randomized controlled trial (RCT) in which students are randomly assigned to receive tutoring (treatment) or not (control). Randomization ensures that the two groups are similar on average, so differences in outcomes can be attributed to tutoring, and a simple comparison of mean exam scores yields an estimate of the causal effect.

However, RCT data are not always available. Participation in tutoring may be voluntary, and students in the treatment and control groups may differ in academic motivation, prior achievement, socioeconomic status, or extracurricular involvement. Moreover, the dataset may not contain measurements on all confounders, such as motivation. In such cases, it is necessary to account for unobserved confounders. Other data scenarios may also arise, requiring different analytical approaches. We summarize them in Table 2.

Framing a causal inference problem involves several steps. One must identify appropriate treatment and outcome vari-

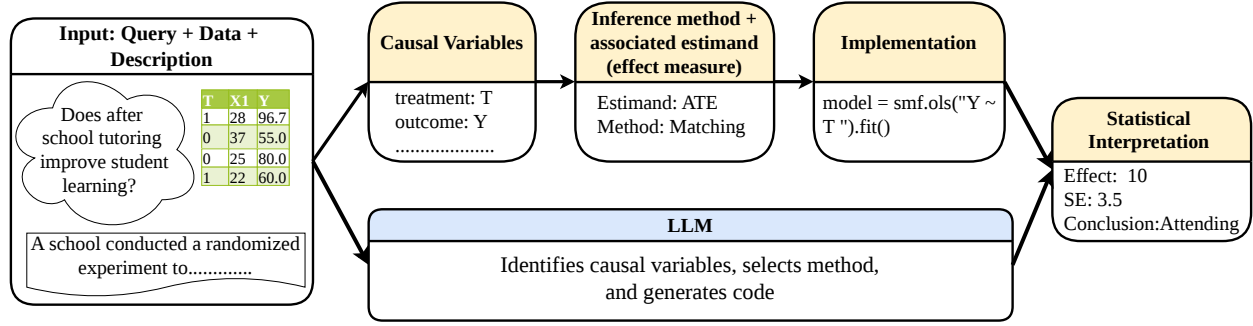


Figure 1. An LLM-automated causal inference pipeline. The top row illustrates the key steps in causal inference: given an input query, data, and study description, one identifies causal variables, selects an estimand and inference method, implements the analysis, and interprets the results statistically. The bottom row shows how we employ an LLM to automate the process.

ables, reason about plausible confounders, select an appropriate method, and implement it to estimate the effect of interest. This process, which combines domain knowledge, causal inference expertise, and coding, has motivated the use of large language models for automating causal inference (Alaa et al., 2024; Kiciman et al., 2024; Chen et al., 2025a; Wang et al., 2025). As LLM-powered tools are increasingly used to support empirical research, understanding their reliability and limitations in causal inference is essential for ensuring that findings are valid. However, no benchmark tests the ability of LLMs to handle the full estimation pipeline, from variable selection to method choice to implementation, particularly in the context of real-world research settings.

Existing benchmarks at the intersection of data-driven causality and LLMs focus on assessing the presence of causal relationships between variables (Mooij et al., 2016; Tu et al., 2019; Huang et al., 2021). Classical benchmark datasets for estimation, such as Jobs (Shalit et al., 2017) and IHDP (Hill, 2011), use simplified settings with fully numerical columns and little variable redundancy. Real-world datasets, especially those derived from surveys, often contain text and redundant information requiring preprocessing before implementation. QRData (Liu et al., 2024) evaluates the implementation of specified causal methods and variables from textbook-style problems, but does not address the selection of appropriate variables and methods, which is a major part of causal inference.

To address these gaps, we present CauSciBench, a benchmark for evaluating the ability of LLMs to perform causal inference for empirical research questions. CauSciBench tests whether LLM agents can (i) select causal variables and an inference method from a scientific question, (ii) implement the method end-to-end on a dataset, and (iii) compute uncertainty and interpret the results statistically. The benchmark draws on three sources: (i) real-world publications, (ii) QRData (Liu et al., 2024), and (iii) synthetic scenarios, allowing us to assess performance across tasks

of varying difficulty. Our evaluation of seven frontier models finds that performance drops substantially on real-world datasets compared to synthetic and textbook settings. This behavior is primarily driven by poor method selection and poor choice of control variables.

2. Related Work

LLMs and causality. Several benchmarks evaluate LLMs’ causal (Jin et al., 2024; Romanou et al., 2023; Nie et al., 2023; Tu et al., 2024) and counterfactual (Chen et al., 2024; Jin et al., 2023; Chen et al., 2025b) reasoning abilities. However, they focus on causal reasoning via text rather than generating causality-driven findings from data. Data-driven benchmarks have focused on causal discovery (Chevalley et al., 2025; Cheng et al., 2024; Zhou et al., 2024), where the goal is to learn causal graphs rather than build causal estimation models to answer questions of interest¹. Meanwhile, benchmarks like QRData (Liu et al., 2024) and RealCause (Neal et al., 2021) evaluate estimation capabilities. However, unlike CauSciBench, they require specifications of models and variables. Hence, they do not test the ability to autonomously select appropriate methods from scratch based on the dataset and its description. CausalBench (Wang, 2024) includes causal effect estimation tasks in mostly synthetic settings using graph-based approaches, and thus does not cover the observational methods used in real-world empirical research. In contrast, CauSciBench builds on real-world causal problems and evaluates the full pipeline from variable selection to method choice to implementation.

LLM Benchmarks for Scientific Discovery. Existing benchmarks evaluate key components of scientific research, including LLMs’ ability to perform statistical rea-

¹It is possible to estimate causal effects from the causal graph using methods like backdoor or frontdoor criterion. In our case, however, we focus on potential outcomes framework which takes a different perspective to causal effect estimation.

Table 1. Comparison of CauSciBench with related benchmark datasets for causal inference. ✓ = Yes, ✗ = No, ~ = Partial. “End-to-End Causal Analysis” indicates whether the benchmark evaluates the full pipeline of causal inference; “Intermediate Evaluation” captures whether the benchmark supports evaluation of intermediate steps; “Data + Context Understanding” assesses whether the benchmark requires models to interpret the relationship between the data variables and the background information. † RealCause uses the Cartesian product of 3 datasets, 4 estimators, 15 ML models, and 10 different hyperparameter settings.

Benchmark	End-to-End Causal Analysis	Intermediate Evaluation	Data + Context Understanding	Sources	Answer Format	# Queries
RealCause (Neal et al., 2021)	✗	✗	✗	3 Datasets + Semi-synthetic	Point Estimate	1569 [†]
QRData (Liu et al., 2024)	✗	~	✓	5 Datasets + 3 Textbooks	Freeform QA	411
DiscoveryBench (Majumder et al., 2025)	✗	✗	✓	26 Datasets + Synthetic	Freeform QA	239
BLADE (Gu et al., 2024)	✗	✓	✓	12 Datasets	Code + Freeform QA	12
CauSciBench	✓	✓	✓	100 Datasets + 2 Textbooks + Synthetic	Point Estimate + Causal Components	367

soning and data analysis (Hu et al., 2024; Wu et al., 2024; Jing et al., 2025). More recent efforts target specialized domains such as machine learning (Huang et al., 2024; Nathani et al., 2025), biology (Laurent et al., 2024), and natural sciences (Chen et al., 2025c). However, these benchmarks do not involve causal analysis, and the social sciences remain underrepresented. BLADE (Gu et al., 2024) and DiscoveryBench (Majumder et al., 2025) introduce open-ended social science problems, but emphasize hypothesis validation with general data science tools rather than rigorous causal inference. In contrast, CauSciBench directly evaluates LLMs on causality-driven analysis across domains, such as healthcare, economics, and political science. Concurrently, Sawarni et al. (2026) has proposed the Causal Reasoning Benchmark for assessing LLMs’ capabilities in performing end-to-end causal inference autonomously. It follows the same design as CauSciBench. However, the two differ in terms of dataset coverage and analysis, with CauSciBench testing a wider range of prompting strategies and frontier models.

3. Problem Setup and Task Definition

We consider a setting in which an agent receives a dataset, a natural language question concerning the effect of one variable on another, and a dataset description that includes the variable definitions, the collection mechanism, and background information. For example, returning to our motivating example on the effect of after-school tutoring on academic performance, the question may be phrased as: *What is the effect of tutoring on academic performance?* or *Does attending after-school tutoring affect academic performance?* While the wording differs, the core task remains the same: estimate the effect of tutoring (treatment) on aca-

demic performance (outcome). Accordingly, how we interpret the results differs. For instance, the first question focuses on the magnitude of the effect, while the second focuses on the statistical significance of the result.

3.1. Input Format

Each problem instance consists of:

- **Dataset:** An experimental or observational dataset
- **Description:** The data collection mechanism, source, and definitions of all variables
- **Query:** A natural-language query asking about the causal effect of one variable on another

3.2. Task

Given this input, the agent must frame and implement an appropriate causal estimation procedure:

- Identify the treatment/outcome/confounding variables
- Select & justify an appropriate causal inference method
- Implement the method to estimate the causal effect
- Interpret the result to answer the query

Accordingly, the LLM output includes the chosen method, the identified variables, runnable implementation code, and the estimated effect with its statistical uncertainty and interpretation.

4. Benchmark Construction

We curate our data from three sources.

Real-world papers. We use datasets from published stud-

Table 2. The choice of causal inference method depends on the data scenario and available information. We illustrate this using different scenarios that may arise when **estimating the effect of after-school tutoring on academic performance**, along with the appropriate causal inference method for each. The "Share" column reports the proportion of CauSciBench instances using each method; Difference-in-Means is subsumed under OLS in the benchmark. One can compute difference-in-means estimate by simply regressing outcome on treatment. Note that these are simplified descriptions; in practice, method selection requires rigorous justification through domain arguments and appropriate statistical tests.

Scenario	Method	Why This Method?	Share
Students in grade 4 were randomly assigned to a tutoring group or a control group, and attendance was mandatory.	Difference-in-Means	Random assignment ensures both groups are similar on average, so any difference in outcomes is due to tutoring.	(OLS)
Same as above, but additional student information such as prior grades and demographics was also collected.	OLS with Controls	Adding student characteristics reduces noise and improves the precision of the estimate.	30.6%
A school ran an optional tutoring program. Data on exam scores and background factors such as prior grades and socioeconomic status were collected.	Propensity Score Methods (Matching or IPW)	Since participation was optional , we use background factors to make students who attended and those who did not comparable, then estimate the effect.	11.9%
School A introduced a tutoring program while School B did not. Exam scores were recorded across multiple years for both schools.	Difference-in-Differences	Over time, many other factors affecting scores could change. By tracking how scores evolved over multiple years in School B (which had no program) , we can remove these changes and isolate the effect of tutoring in School A.	18.0%
A school decided to offer a tutoring program. Students who scored below a cutoff of 40 were enrolled in the program.	Regression discontinuity	Students just above and below the cutoff could be very similar in terms of pre-treatment variables. Any difference in their outcomes can be attributed to receiving tutoring.	12.0%
Students were randomly selected to be offered tutoring , but participation was optional . Not all selected students attended.	Instrumental Variables	Being offered tutoring is not the same as attending. Since the offer was random, we use it as an instrument to isolate the effect of <i>actually attending</i> the program, which was optional .	16.9%
Students were assigned to tutoring but could attend for any duration. Data on hours attended was recorded, but information on all factors influencing attendance was not fully collected.	Frontdoor Estimator	The full effect of the program runs through the hours attended (the mediator). Tracking this path allows us to estimate the causal effect even without observing all confounders.	4.0%

ies, which capture the complexity of real-world data: large variable sets, noise, and messiness. Reference answers (method choice and causal effect estimates) are directly from the paper authors, ensuring they are well justified.

QRData. We include causal inference examples from QRData (Liu et al., 2024), a benchmark designed to evaluate LLMs’ broader causal reasoning skills. The causal inference examples, which we term **QRData-CI**, are drawn from standard causal inference datasets and textbooks. Textbook datasets are largely designed for pedagogical purposes and are preprocessed to streamline implementation. Similarly, standard datasets require minimal preprocessing because they are designed to evaluate new estimators.

Synthetic scenarios. Synthetic data allows us to specify the ground-truth causal effect in a controlled setting, removing reliance on reference answers from external sources. Moreover, this mirrors common practice in causal estimation research, where new methods are evaluated on synthetic data with known causal effects. The three

sources enable comparative analysis across difficulty levels, thereby allowing us to identify gaps and opportunities for LLM-powered causal inference.

4.1. Instance Creation from Published Studies

We construct benchmark instances from published empirical studies in causal inference, selected by the authors, who have a background in statistics and causal inference. Papers were selected based on three criteria: (1) the dataset must be publicly available (e.g., Harvard Dataverse, Yale ISPS, R package repositories), (2) the paper must be published, and (3) reliance on a particular method to convey findings. The benchmark instances describe only the dataset, excluding any reference to the original study name, method used, or its findings. Additionally, we run tests to rule out data contamination. The procedure is described in Section H of the appendix.

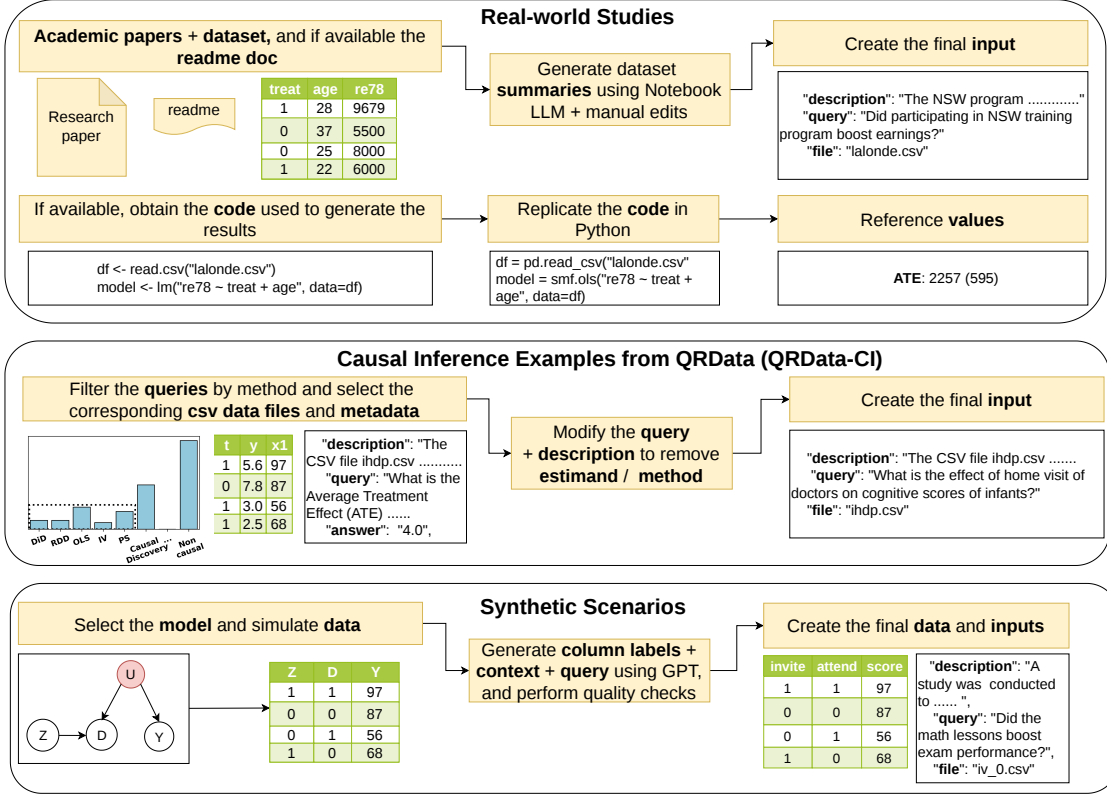


Figure 2. Illustration for building the benchmark using three sources: Real-world papers, QRData, and synthetic scenarios.

4.2. Annotation

Annotations were performed by the authors based directly on the published studies. We examined the dataset, main empirical results, and reported causal estimates to identify the primary research question, which forms the query for each instance. A sample annotation is shown in Figure 3.

Reference answers are taken from the reported results, including the method, effect estimate, and standard errors. Since causal inference relies on domain-specific assumptions with no universally correct approach, we treat the original authors as experts and use their analysis as the reference. Annotators also replicate the results in Python to verify that the reference answers are attainable.

After the initial annotation, two authors reviewed each instance to ensure consistency in annotation format and to check that inputs did not reveal the method, findings, or variables (e.g., treatment, outcome). Because annotations are grounded directly in published results and each result uses a single primary method, there was little ambiguity.

4.3. Construction of QRData-CI and Synthetic Data

QRData-CI. QRData (Liu et al., 2025) contains causal inference tasks from causality textbooks (Alves, 2022; Imai, 2018) and common causality benchmarks, including Jobs (Shalit et al., 2017) and IHDP (Hill, 2011). We selected causal inference queries with numerical answers. Similarly, we omitted queries that require methods outside the scope of CauSciBench. For instance, those involving machine learning methods or more advanced techniques such as synthetic control. While QRData specifies the inference method or estimand, we remove any explicit references to causal estimation methods. As an example, the query related to the IHDP dataset (Hill, 2011), *What is the Average Treatment Effect (ATE) of the dataset?*, becomes *What is the effect of home visits by doctors on cognitive scores of infants?*

Synthetic Scenarios. We randomly select the true causal effect $\tau \in (1, 10)$. Continuous covariates are drawn from a normal distribution, while binary covariates and treatment assignments (for binary treatment settings) are generated from a binomial distribution. The outcome Y is determined by the model specification. For example, in the case of a randomized trial: $Y = \alpha + X\vec{\theta} + \tau T + \epsilon$, where $\epsilon \sim \mathcal{N}(0, 1)$ is the error term, $\vec{\theta} \sim \mathcal{N}(u, kI)$, and α is

Sample Query Based on Card (1993)

Paper Source: Using geographic variation in college proximity to estimate the return to schooling.

Description: The National Longitudinal Survey of Young Men (NLSYM) was conducted to collect data on demographics, education, and employment outcomes. Participants were tracked over time to study long-term patterns. The dataset used here comes from the 1976 wave of the survey. Variables in the dataset:

- **lwage:** log of wages
- **educ:** years of education
- **exper:** years of work experience
- **black:** 1 if Black, 0 otherwise
- **south:** 1 if lives in a southern state, 0 otherwise
- **married:** 1 if married, 0 otherwise
- **smsa:** 1 if living in a metropolitan area, 0 otherwise
- **nearc4:** 1 if there is a four-year college in the county, 0 otherwise

Query: What is the effect of education on earnings?

Answer: 0.132

Standard Error: 0.049

Is Significant: 1

Method: IV (Instrumental Variable)

Instrument Variable: nearc4

Dataset File: card_geographic.csv

Reference in Paper: Table 4

Field: Economics

Figure 3. Sample benchmark instance with color-coded **treatment variable**, **outcome variable**, and **control covariates**, along with other attributes, including the method (IV), the dataset filename, and the corresponding reference in the source paper.

the intercept. Similarly, X denotes the controls and T is the treatment. We prompted GPT-4o to synthesize diverse plausible scenarios explaining how and why the data were collected. Accordingly, the LLM also produces metadata, such as headings and descriptions for the covariates, treatment variables, and outcomes. The prompt template used for this task, along with an example of the generated context and query, is provided in Appendix C.

4.4. Method Distribution

In our benchmark, we use examples associated with the following methods: regression discontinuity design (RDD) (Imbens & Lemieux, 2008), instrumental variables (IV) (Imbens, 2014), ordinary least squares (OLS) (Huntington-Klein, 2021; Cunningham, 2021), difference-in-differences (DiD) (Roth et al., 2023), matching methods (Stuart, 2010), propensity score-based methods (PS) (Rosenbaum & Rubin, 1983; Austin, 2011), generalized linear models (GLMs) (Breen et al., 2018), as well as frontdoor adjustment (FD) (Pearl, 2009). These methods are used actively in causal inference research. The as-

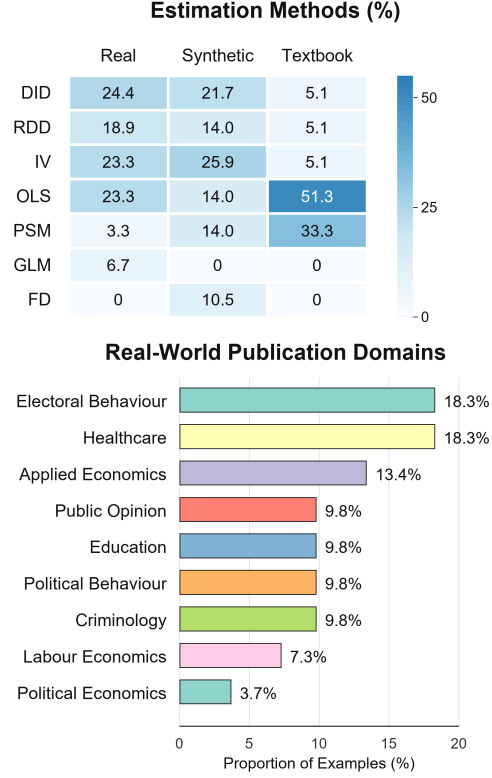


Figure 4. Distribution of paper domains in real-world publications and causal effect estimation across the three dataset collections.

sumptions underlying the methods differ, which dictate the scenarios in which they can be applied. We describe the methods in detail in Appendix Section I.

5. Experimental Setup

We evaluate 4 different prompting strategies: Direct Prompting (Brown et al., 2020), Chain of Thought (CoT) (Wei et al., 2022), Program of Thought (PoT), and ReAct (Chen et al., 2023). We adapt the base templates from (Liu et al., 2024) and modify them to enable end-to-end inference. The detailed prompt templates are provided in Appendix B.

Implementation. For causal effect estimation, we use the DoWhy (Sharma & Kiciman, 2020; Blöbaum et al., 2024), linearmodels (Sheppard et al., 2024), rdd (Magnusson, 2019), and statsmodels (Seabold & Perktold, 2010) libraries. Similarly, for preprocessing and intermediate computations, we use numpy (Harris et al., 2020), pandas (pandas development team, 2020), and scikit-learn (Pedregosa et al., 2011). Additionally, the LLM-generated code is run using Docker containers. Across all models, we allow up to two code-repair retries using the execution error messages.

5.1. Evaluation Metrics

Let N denote the total number of examples for which results were extracted. We evaluate all models using the following metrics:

(1) Method Correctness (MC): Macro-averaged F1 score measuring whether the LLM-selected causal method ($\hat{m}_i$) matches the reference method (m_i), where M is the number of method classes and P_m and R_m denote the precision and recall for method class m . We use macro-averaged F1 to account for class imbalance across causal inference methods.

$$MC = \frac{1}{M} \sum_{m=1}^M \frac{2 \cdot P_m \cdot R_m}{P_m + R_m} \times 100\% \quad (1)$$

(2) Effect Accuracy (EA): Percentage of queries for which the LLM-generated answer ($\hat{\tau}_i$) is within 5% of the reference answer (τ_i).

$$EA = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[\left| \frac{\hat{\tau}_i - \tau_i}{\tau_i} \right| \leq 0.05 \right] \times 100\% \quad (2)$$

Minor differences, such as the choice of control variables, may not lead to exact answers, so we consider any result within 5% of the reference answer to be correct. A similar metric has been used in Liu et al. (2024).

(3) Variable Selection Accuracy (VSA): The percentage of queries for which the LLM-selected treatment/outcome/controls/confounders ($\hat{V}_i$) match the reference values (V_i).

$$VSA = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{V}_i \cap V_i|}{|V_i|} \times 100\% \quad (3)$$

Additionally, we compute method selection accuracy and the relative errors of the causal effects. The corresponding results are provided in Table 6 of the appendix.

6. Results and Discussion

We tested seven frontier models from leading model families, including OpenAI, Gemini, Grok, and Qwen. Table 3 presents the results across all models and prompting strategies for the three dataset collections. A comparison of performance shows that both method correctness and effect accuracy consistently lag behind those achieved on the synthetic and QRData datasets across all strategies. On average, method correctness reaches 45.8% on the real-world datasets, compared to 69.9% and 63.5% on the synthetic and QRData-CI datasets, respectively. The gap is even larger for effect accuracy, where performance drops to 14.4% on the real-world datasets, compared to 53% and 30.5% on the synthetic and QRData datasets. These results highlight the challenges of causal estimation from

(gpt-4o-mini)							
Reference Method	DiD	GLM	IV	NA	OLS	PS	RDD
	4	2	1	0	29	0	0
	0	0	0	1	9	0	0
	0	1	0	1	41	0	0
	0	0	0	0	0	0	0
	0	3	0	4	53	0	0
	0	0	0	1	11	0	0
	2	1	0	0	20	0	1
Predicted Method							

(gpt-4o)							
Reference Method	DiD	GLM	IV	NA	OLS	PS	RDD
	21	1	4	1	9	0	0
	0	2	1	0	7	0	0
	1	1	16	0	23	1	1
	0	0	0	0	0	0	0
	2	2	4	0	48	4	0
	0	0	0	0	7	5	0
	2	1	1	0	7	1	12
Predicted Method							

Figure 5. Confusion matrix for method selection across the three datasets for GPT-4o-mini and GPT-4o for direct prompting. NA indicates failure to extract a suitable method. The red boxes highlight the over-reliance of models on ordinary least squares (OLS). However, this over-reliance is reduced for larger models.

real-world data, both in selecting the appropriate method and in implementing it correctly.

There are several reasons for this. Real-world scenarios are inherently more complex, particularly when the data are observational. Performing causal estimation in such settings requires stronger reasoning about identification assumptions and causal structure. In addition, the datasets often contain more variables, and variable names can be similar, making it more difficult to identify the relevant ones. In contrast, the QRData and synthetic datasets are generally more well-defined and structured, which simplifies the inference process.

6.1. Fine-grained error analysis

To identify the major bottlenecks in performing causal inference using LLMs, we perform a fine-grained analysis.

Table 3. Comparison of method correctness ($\uparrow$) and effect accuracy ($\uparrow$) across datasets, models, and prompting strategies. For each model, **dark green** indicates the highest method F1 and **dark red** indicates the highest effect accuracy across the 4 prompting approaches.

Dataset	Model	Method Correctness ($\uparrow$)				Effect Accuracy ($\uparrow$)			
		Direct	CoT	PoT	ReAct	Direct	CoT	PoT	ReAct
Real	Gemini-2.5-Flash-Lite	30.83	44.10	39.67	42.60	12.05	12.43	5.96	9.55
	Grok-4-Fast	59.24	62.51	41.08	44.23	18.07	25.00	21.94	13.71
	GPT-5-mini	60.58	53.75	49.64	51.84	13.41	25.33	21.58	21.88
	GPT-4o-mini	12.57	15.41	26.78	16.90	5.16	4.49	5.45	9.89
	GPT-4o	51.12	51.92	40.97	41.99	10.90	10.84	7.56	14.63
	OpenAI-o3	65.74	68.48	51.42	72.60	24.52	23.53	18.49	12.20
	Qwen3-Next-80B-Inst	44.56	49.20	51.41	40.14	18.02	12.21	12.35	10.84
QRData-CI	Gemini-2.5-Flash-Lite	74.67	77.01	64.50	75.02	26.32	28.95	23.08	28.21
	Grok-4-Fast	62.28	75.09	80.10	74.4	33.33	36.84	35.90	34.21
	GPT-5-mini	85.44	88.39	71.79	82.49	33.33	32.43	20.59	41.18
	GPT-4o-mini	13.56	48.15	46.72	27.15	27.78	31.43	40.00	30.30
	GPT-4o	59.17	84.18	80.10	72.38	23.08	23.68	47.22	27.78
	OpenAI-o3	84.24	77.86	84.24	79.23	41.03	24.00	30.77	0.00
	Qwen3-Next-80B-Inst	70.74	72.85	74.94	70.74	35.90	28.21	33.33	34.21
Synthetic	Gemini-2.5-Flash-Lite	67.99	83.52	69.45	60.34	35.82	47.29	34.09	33.81
	Grok-4-Fast	68.63	69.51	59.27	63.86	68.33	67.19	62.12	48.18
	GPT-5-mini	88.99	91.60	80.86	65.78	68.50	67.21	64.00	69.37
	GPT-4o-mini	7.27	15.75	16.92	15.91	58.87	54.55	47.01	56.64
	GPT-4o	64.0	82.11	78.58	54.20	52.21	50.72	54.33	55.47
	OpenAI-o3	73.71	91.56	75.46	68.39	71.09	40.28	62.96	7.14
	Qwen3-Next-80B-Inst	56.18	70.14	78.55	60.36	59.85	51.09	50.70	44.78

Below we focus primarily on the Real dataset.

Method selection and control variable selection are the major bottlenecks. While outcome and treatment identification (Table 4) achieve relatively high accuracy, the overlap scores for model covariates are consistently lower across all dataset types, especially for real datasets. This disparity suggests that models can identify the primary causal variables with reasonable success but struggle to determine the appropriate set of controls and confounders required for valid causal estimation. A key reason for this difficulty is the large number of variables present in real datasets, which makes it challenging for models to systematically evaluate and select the relevant subset. While for data from randomized trials the choice of control variables does not affect the results significantly, for observational studies, the results are more sensitive to the choice of variables.

Wrong methods directly amplify estimation errors. Table 4 shows that incorrect method selection is the primary driver of causal inference failures, yielding lower effect accuracy in most settings. Errors in method selection propagate through subsequent stages of the causal inference pipeline, as the statistical model varies across methods. The textbook-based dataset (Table 8) is an exception: most misclassifications involve choosing IPW over difference-in-means for variants of the IHDP dataset (Hill, 2011). The data come from a randomized trial, so both methods yield similar causal effects, and the resulting error remains small.

In contrast, for observational datasets in the collection, the wrong choice of controls or confounders can yield larger errors despite selecting the correct method.

Models systematically default to OLS estimation. The confusion matrices in Figure 5 reveal that LLMs exhibit a pronounced bias toward Ordinary Least Squares (OLS) across all causal inference scenarios, regardless of the reference method. This tendency is particularly pronounced for smaller models, such as GPT-4o-mini. The overwhelming selection of OLS stems from several factors. OLS is simpler and easier to implement. Likewise, for most empirical papers, OLS is the baseline model. However, this bias is highly problematic for causal inference. Naive OLS often fails to account for the effects of unobserved confounders. Hence, when possible, researchers use instrumental variables. Likewise, even when confounders are observed, naive OLS-based estimates exhibit low precision. Thus, practitioners often employ techniques like matching (Dehejia & Wahba, 2002).

Pass@k partially mitigates but does not resolve method selection failures. All main results are based on a single trial (Pass@1). To assess robustness to output stochasticity, we ran three independent trials (Pass@3) for GPT-4o and GPT-4o-mini (Appendix G). Pass@3 improves method selection accuracy by 7.3 pp on average, with gains ranging from 0 to 25.7 pp depending on the setting. These modest gains suggest that most failures reflect systematic limitations in causal reasoning beyond stochastic variation.

Table 4. Performance of different models on the Real dataset averaged across all prompting strategies. The first block presents accuracies for key causal quantities: method and variable selection. Meanwhile, the second block presents causal effect accuracy for cases in which the correct method is selected versus when an incorrect method is selected. Red values indicate, for each model, the lowest-performing metric in its row, highlighting the area where that model shows its greatest relative weakness. The results show that control variable overlap and incorrect method selection are the primary drivers of errors. Abbreviations: MCF = method correctness F1 score; TVSA = treatment variable selection accuracy; OVSA = outcome variable selection accuracy; CVSA = control variable selection accuracy; EA = effect accuracy.

Model	Method & Variable Selection				Causal Effect Computation		
	MCF	TVSA	OVSA	CVSA	EA (correct method)	EA (wrong method)	Δ EA
Gemini 2.5 Flash-Lite	39.30	76.72	95.67	47.08	15.83	4.20	11.63
GPT-4o	46.50	77.66	95.00	49.07	17.03	2.54	14.49
GPT-4o mini	17.91	83.90	92.62	48.65	12.88	2.66	10.22
GPT-5 mini	53.95	78.65	94.78	55.31	26.63	4.70	21.93
Grok 4-Fast	51.76	79.46	96.37	39.62	25.30	6.56	18.74
OpenAI-o3	64.56	82.95	94.67	51.86	25.70	3.40	22.30
Qwen3-Next-80B-Inst	46.32	73.90	94.91	50.43	17.95	7.06	10.89

6.2. Comparison of prompting strategies

There is no dominant prompting strategy for causal analysis, as shown in Table 6. CoT generally performs well in method selection because it encodes the causal analysis pipeline and reasoning about identification assumptions. PoT is more implementation-focused and less effective at reasoning about untestable assumptions. ReAct, while free-form with less guidance, shows smaller overlap in control/confounders (Table 10) but is the most efficient in implementation, requiring fewer code generation attempts. Our findings differ from those of Liu et al. (2024), likely because their benchmark emphasizes implementation tasks where ReAct and PoT excel, whereas CauSciBench involves reasoning followed by implementation.

Reasoning-heavy prompts affect coding. Table 7 reports the coding success rates, i.e., the rate at which a model produces executable code, across datasets, prompt types, and models. Overall, most models achieve high success rates, averaging 90.5% when excluding o3, which is a notable exception. Its success rate falls sharply under CoT and ReAct prompting. For instance, they drop to 37.3% and 22.2%, respectively, on real data. This is perhaps because the reasoning structure imposed by these prompts conflicts with the model’s internal reasoning, leading it to produce incomplete outputs.

7. Conclusion

We present **CauSciBench**, a benchmark for LLM-powered causal inference on tasks resembling real-world research. Our findings demonstrate that frontier LLMs exhibit a systematic bias toward methodological oversimplification (often defaulting to OLS) and struggle with implementation

(leading to many code-generation attempts). Moreover, the performance gap between synthetic and real-world scenarios highlights several areas for advancing LLM-based causal inference. First, there is a need for datasets that better capture real-world complexity, including noise, structure (numeric vs. non-numeric), and variable dimensionality. Second, there is a need for frameworks that enable LLMs to reason carefully about method assumptions and support their implementation. Based on our observations, we conclude that developing frameworks that combine CoT’s reasoning capabilities with ReAct’s implementation capabilities could be promising.

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Impact Statement

This paper presents work aimed at advancing the field of AI-enhanced causal inference, which facilitates empirical, data-driven research across many disciplines. Currently, we do not see any potentially harmful social implications from this work.

Limitations

The results reported are primarily based on pass@1. We acknowledge the stochastic nature of LLM outputs, and pass@k would lead to more stable results. We ran pass@3 on two OpenAI models. However, due to budgetary constraints, we were unable to run experiments on other models. Causal inference is a broad topic. The two schools of thought are Pearl’s structural modeling approach based on causal graphs and the Potential Outcomes Framework, also called the Neyman-Rubin framework. Our focus is on the latter, since it is the dominant paradigm for causal effect estimation. Likewise, our focus is mostly on tabular datasets, as this is the most common data form in empirical research. We have not included tasks that involve inferring causal effects from text data. This is an up-and-coming research topic accelerated by the recent advancements in machine learning and natural language processing. Likewise, there have been several recent advancements in estimation methods leveraging machine learning. One popular example is Double Machine Learning (DML). Since our benchmark is the first of its kind, we currently focus on foundational methods. These are still actively used in day-to-day research.

The benchmark does not test several aspects of causality, including assumption validation, sensitivity analysis, and robustness tests. Quantitatively evaluating these is challenging. They are often specific to the scenario at hand.

For instance, one cannot predetermine the alternative treatment year for a placebo test in difference-in-differences. We make the decision based on the available data. Similarly, evaluating the justification of assumptions objectively is challenging. For instance, the explanation for the plausibility of exclusion restriction for an instrument relies on domain knowledge. The choice and justifications given by an LLM may vary from those in the reference. Similarly, the parallel trends in difference-in-differences require visual inspection. One can employ tools such as LLM-as-a-judge, but the benchmark cannot support this. Hence, given the constraints, we focus on evaluating the choice of inference method currently.

Declaration of LLM Usage

We have used commercially available LLMs to proofread our draft and fix grammatical errors. Similarly, when constructing the benchmark, the authors used Notebook LLM to gather key details about the dataset from the respective papers, as relevant information may be scattered across the papers. Likewise, we used commercial LLMs to proofread and fix grammatical errors in the dataset description component of the annotations that we originally crafted.

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A. Sample Questions From Each Source

Real-World Publications
Source: Cities as Lobbyists (Goldstein & You, 2017)
Domain: Economics
Natural Language Query: How much does the money spent on lobbying increase the number of earmarks received? Method: Instrumental Variables Treatment: <code>ln_citylobby</code> (log of city lobbying spending) Instrument: <code>direct_flight_dc</code> (1=direct flight to DC in 2007, 0=otherwise) Outcome: <code>ln_earmark</code> (log of total earmarks 2008-2009) Other Variables: <code>state</code> , <code>pop_e</code> , <code>land_e</code> , <code>water_e</code> , <code>senior_e</code> , <code>student_e</code> , <code>ethnic_e</code> , <code>mincome_e</code> , <code>unemp_e</code> , <code>poverty_e</code> , <code>gini_e</code> , <code>city_propertytaxshare_e</code> , <code>city_intgovrevenue</code> , <code>share_e</code> , <code>city_airexp_e</code> , <code>houdem_e</code> , <code>ln_countylobby</code> Data: Cities with population over 25,000, 2007-2009 panel
Synthetic Dataset
Source: Cardiovascular Rehabilitation Program Effectiveness Study
Domain: Healthcare
Natural Language Query: Does the new rehabilitation program help patients with cardiovascular diseases recover faster? Method: Regression Discontinuity Design Treatment: <code>treatment_received</code> (1=new program, 0=standard care) Running Variable: <code>income_level</code> (threshold at 12 for eligibility) Outcome: <code>recovery_time</code> (days to recovery) Other Variables: <code>patient_age</code> , <code>health_index</code> , <code>smoking_status</code> , <code>obesity_status</code> Data: Regional health department evaluation study
Textbook Examples
Source: Effect of Cigarette Taxation on Consumption (Liu et al., 2024)
Domain: Healthcare, Political Science
Natural Language Query: Did Proposition 99 help reduce cigarette sales? Method: Difference-in-Differences Treatment: <code>california</code> (1=CA with Prop 99, 0=other states) Time: <code>after_treatment</code> (1=post-1988, 0=pre-1988) Outcome: <code>cigsale</code> (total cigarette sales) Other Variables: <code>state</code> , <code>year</code> , <code>lnincome</code> , <code>beer</code> , <code>age15to24</code> , <code>retprice</code> Data: 39 US states, 1970-2000 panel

Table 5. Sample questions from each source pillar with the information regarding the paper that the LLM uses as context.

B. Prompt Templates

In this section, we present the templates for the four prompting methods: **Direct Prompt**, **Chain of Thoughts (CoT) prompt**, **Program of Thoughts (PoT) prompt**, and **ReAct**

Direct Prompt

You are an expert in statistics and causal reasoning. You will answer a causal question on a tabular dataset.

The dataset is located at `{self.dataset_path}`.

The dataset has the following description: `{self.dataset_description}`

To help you understand it, here is the result of `df.describe()`:

`{df_info}`

Here are the columns and their types:

`{columns_and_types}`

Here are the first 5 rows of the dataset:

`{df.head() }`

If there are fewer than 10 columns, here is the result of `df.cov()`:

`{(df.cov(numeric_only=True) if len(df.columns) < 10
else "Too many columns to compute covariance")}`

Here is the output of `df.isnull().sum(axis=0)`: `{nan_per_column}`

The causal question I would like you to answer is: `{self.query}`

Using the descriptions and information from the dataset, write Python code to build the causal inference model based on the method and variables you have selected, and compute the causal effect to answer the query. If you need to preprocess the data, please do so in the code.

Important: Only use these approved packages: pandas, numpy, scipy, scikit-learn (sklearn), statsmodels, dowhy, rdd (for regression discontinuity design), linearmodels, econml.

Here are some example methods; you can choose one from them:

- IPW (Inverse Probability Weighting): choose the right estimand (ATE/ATT/ATC), and compute the causal effect
- Linear regression with control variables: build a regression model with the treatment, outcome, and confounders/control variables, and compute the causal effects
- Instrumental variable: build an instrumental variable model, and compute the causal effects associated with the treatment variable
- Matching: choose the correct estimand (ATE/ATT/ATC), and match accordingly, and then compute the causal effects
- Difference-in-differences: build a difference-in-differences model, and output the coefficient of the variable of interest
- Regression discontinuity design: build a regression discontinuity design model, and output the coefficient of the variable of interest
- Linear regression / difference-in-means: either build a regression model consisting of the treatment and outcome variables, and compute the coefficient associated with the treatment variable or compute the difference in means across treatment and control groups
- Generalized linear models / GLM: build a GLM model, and output the coefficient of the variable of interest
- Frontdoor adjustment: build a causal graph, identify a mediator variable between the treatment and outcome, check for frontdoor criterion, and compute the causal effect using the frontdoor adjustment formula

Make sure the code prints the final results, including:

1. The causal effect (the value only)
2. The standard deviation (the value only)
3. The causal inference method that was used to compute the effect (the method name only)
4. The treatment variable (the variable name only)

5. The outcome variable (the variable name only)
6. The mediator variable (the variable name only if frontdoor adjustment was used)
7. RCT: True / False (NA if not sure; whether the data is from a randomized controlled trial or not)
8. The covariates / control variables that were used in the causal inference model (the variable names only)
9. Instrumental variable, if instrumental variable method was used (the variable name only)
10. Running variable, if regression discontinuity design was used (the variable name only)
11. Temporal variable, if difference-in-differences was used (the variable name only)
12. Results of statistical tests, if applicable
13. Brief explanation for model choice
14. The regression formula, if applicable.

If a variable is not applicable, print "NA" for it.

The code you output will be executed, and you will receive the output. Please make sure to output only one block of code, and make sure the code prints the result you are looking for at the end. Everything between your first code block: ```python and ``` will be executed. If there is an error, you will have several attempts to correct the code.

Chain of Thoughts Prompt

You are an expert in causal inference. You will use a chain-of-thought approach to answer a causal question on a tabular dataset.

The dataset is located at {self.dataset_path}

The dataset has the following description: {self.dataset_description}

Here are the columns and their types: columns_and_types

Here is the statistical summary of the dataset: df.describe()

Here are the first 5 rows of the dataset: {df.head() }

If there are fewer than 10 columns, here is the result of df.cov():

```
{(df.cov(numeric_only=True) if len(df.columns) < 10
else "Too many columns to compute covariance")}
```

Here is the output of df.isnull().sum(axis=0): {nan_per_column}

The causal question I would like you to answer is: {self.query}

Let us approach this problem step by step.

Step 1. First, go through the dataset description and the columns and their types. Then, identify the treatment variable, the outcome variable, and the potential confounders. Explain your reasoning for choosing these variables. Remember, the dataset is located at: {self.dataset_path}.

Step 2. What would be the right estimand to consider for this problem? Then, choose the most appropriate method that can be used to estimate the causal effect. The available methods are:

- IPW (Inverse Probability Weighting): Choose the right estimand (ATE/ATT/ATC), and compute the causal effect
- Linear regression with control variables: Build a regression model with the treatment, outcome, and confounders/control variables, and compute the causal effects
- Instrumental variable: Build an instrumental variable model, and compute the causal effects associated with the treatment variable
- Matching: Choose the correct estimand (ATE/ATT/ATC), and match accordingly, and then compute the causal effects,
- Difference-in-differences: Build a difference-in-differences model, and output the coefficient of the variable of interest

- Regression discontinuity design: Build a regression discontinuity design model, and output the coefficient of the variable of interest
- Linear regression / difference-in-means: Either build a regression model consisting of the treatment and outcome variables, and compute the coefficient associated with the treatment variable or compute the difference in means across treatment and control groups
- Generalized linear models / GLM: Build a GLM model, and output the coefficient of the variable of interest,
- Frontdoor adjustment: Build a causal graph, identify a mediator variable between the treatment and outcome, check for frontdoor criterion, and compute the causal effect using the frontdoor adjustment formula

Explain why you chose the selected method, and how the data and its description support your choice. This means you should explain why the identification assumptions of the method are satisfied.

Step 3. Next, we will plan the implementation. Before writing the code, describe your implementation process. This includes:

1. Describing the necessary pre-processing steps.
2. How we will select the variables to use in the model?

Step 4. Finally, reflecting on the previous steps, write Python code to answer the causal question: `{self.query}`. Feel free to preprocess the data.

Important: Only use these approved packages: pandas, numpy, scipy, scikit-learn, statsmodels, dowhy, rdd, linearmodels, econml.

Use the methods from the above libraries to implement the method you chose. Be careful about implementation.

Make sure the code prints the final results, including:

1. The causal effect (the value only)
2. The standard deviation (the value only)
3. The causal inference method used (the method name only)
4. RCT: True / False / NA
5. The treatment variable
6. The outcome variable
7. The mediator variable (if applicable)
8. The covariates / control variables
9. Instrumental variable (if applicable)
10. Running variable (if applicable)
11. Temporal variable (if applicable)
12. Results of statistical tests, if applicable
13. Brief explanation for model choice
14. The regression formula, if applicable

If a variable is not applicable, print "NA" for it.

The code you write will be executed, and you will next analyze the output. To ease the process, please output one block of code, and make sure the code prints the key results and values. Everything between your first code block: ````python` and ````` will be executed. If there is an error, you will have several attempts to correct the code. Hence, if there is an error, please fix it and re-run.

Program of Thoughts Prompt

You are a causal inference expert. Your goal is to generate a causality-driven answer to the user query: {self.query} using the provided data.

The description and the query can be found below. Please analyze the input information and write a Python code that performs causal effect estimation.

You can use the following libraries: pandas, numpy, scipy, sklearn, statsmodels, dowhy, rdd, linearmodels, econml.

The format of the code should be:

```
```python
def causal_analysis():
 # import libraries
 # load data
 # identify treatment, outcome, confounders,
 # control variables (pre-treatment variables)
 # select appropriate causal method, and method-specific variables
 # estimate causal effect and standard error
 # print results (12 items listed below). This is important
 # return a dictionary containing the 12 items listed below
result = causal_analysis()
```
```

Available causal inference methods: IPW (Inverse Probability Weighting), Linear regression with control variables, Instrumental variable, Matching, Difference-in-Differences, Regression Discontinuity Design, Linear Regression/Difference-in-Means, Generalized linear models, Frontdoor adjustment.

Print the following 12 items in the code:

1. Causal effect
2. Standard error
3. Method name
4. RCT (True/False/NA)
5. Treatment variable
6. Outcome variable
7. Mediator variable
8. Control covariates used
9. Additional variable
10. Statistical test results
11. Model choice explanation
12. Regression formula (if applicable)

If a field is not applicable, print "NA".

Here is information about the data. Data Description: {self.dataset_description}

Dataset Location: {self.dataset_path}

Columns and types: {columns_and_types}

First 10 rows: {df.head(10)}

Missing values: {nan_per_column}

Likewise, the query is: {self.query}

Everything between your first code block: python and will be executed. If there is an error, you will have several attempts to correct the code.

ReAct Prompt

Data Description: {self.dataset_description}

The dataset is located at {self.dataset_path}

You are a causal inference expert working with a pandas dataframe in Python. The name of the dataframe is 'df'

You should use the tools below to answer the causal question of interest:

'python_repl_ast': A Python shell. Use this to execute Python commands. Input should be a valid Python command.

When using this tool, sometimes output is abbreviated - make sure it does not look abbreviated before using it in your answer.

Important: Only use these approved packages: pandas, numpy, scipy, scikit-learn (sklearn), statsmodels, dowhy, rdd, linearmodels, econml.

Use the following format

Question: The input question you must answer

Thought: Your thoughts on what to do next. You need to think carefully.

Action: The action to take, should be python_repl_ast

Action Input: The input to the action, should be the code to execute

Observation: The result of the action

...(this Thought/Action/Action Input/Observation can repeat N times)

Thought: I now know the final answer

Final Answer: The final answer to the original input question. Please provide a structured response including the following information. If a field is not applicable, use "NA".

- Causal Effect: [The causal effect estimate]
- Method: [The method used]
- Standard Error: [The standard error of the causal effect]
- Treatment Variable: [The treatment variable]
- Outcome Variable: [The outcome variable]
- Mediator Variable: [The mediator variable, if frontdoor adjustment was used, NA otherwise]
- RCT: [True / False indicating if the data is from a randomized controlled trial, NA if not sure]
- Covariates: [List of control covariates and confounders used in the estimation model]
- Additional Variable: [Instrument, running variable, or temporal variable, if applicable]
- Results of Statistical Tests: [Key statistical results, if applicable]
- Explanation for Model Choice: [Explanation, if applicable]
- Regression Formula: [The regression formula, if applicable]

Note: Only import from the approved package list above. Do not use any other packages. Do not create any plotting.

For all outputs in code, THE 'print()' function MUST be called. If you use Action in this step, stop after generating the Action Input and wait the execution outcome from 'python_repl_ast'. If you output the final answer in this step, do not use Action.

Here is an example of using the 'python_repl_ast':

Action: python_repl_ast

Action Input:

```
```python
Your code goes here - only use approved packages
import pandas as pd
import numpy as np
print(df.head())
```
Begin!|
```

Question: self.query

Available causal inference methods:

- IPW (Inverse Probability Weighting): Choose the right estimand (ATE/ATT/ATC), and compute the causal effect

- Linear regression with control variables: Build a regression model with the treatment, outcome, and confounders/control variables, and compute the causal effects
- Instrumental variable: Build an instrumental variable model, and compute the causal effects associated with the treatment variable
- Matching: Choose the correct estimand (ATE/ATT/ATC), and match accordingly, and then compute the causal effects
- Difference-in-differences: Build a difference-in-differences model, and output the coefficient of the variable of interest
- Regression discontinuity design: Build a regression discontinuity design model, and output the coefficient of the variable of interest
- Linear regression / difference-in-means: Either build a regression model consisting of the treatment and outcome variables, and compute the coefficient associated with the treatment variable or compute the difference in means across treatment and control groups
- Generalized linear models / GLM: Build a GLM model, and output the coefficient of the variable of interest
- Frontdoor adjustment: Build a causal graph, identify a mediator variable between the treatment and outcome, check for frontdoor criterion, and compute the causal effect using the frontdoor adjustment formula

C. Synthetic Data Generation

We use the template below to generate the context and variable labels for synthetic data.

Prompt for Generating Context for Synthetic Data

You are a helpful assistant generating realistic, domain-specific contexts for synthetic datasets. The current dataset is designed for {method_name} studies in the domain domain.

Dataset Summary
{summary}

Previously Used Contexts (avoid duplication)
{history}

Domain-Specific Guidance
{domain_guides}

Your Tasks:

1. Propose a realistic real-world scenario that fits a {method_name} study in the domain domain. Mention whether the data was collected from a randomized trial, policy rollout, or real-world observation.
 - a. Assign realistic and concise variable names in snake_case. Map original variable names like "X1" to names like "education_years".
 - b. Provide a one-line natural-language description for each variable (e.g., education_years: total years of formal schooling completed by the individual.). Use newline-separated key-value format.
2. Write a paragraph describing the dataset's background: who collected it, what was studied, why, and how. Then, provide a clear description of each variable in the dataset, explaining what it represents and, where relevant, its type (e.g., continuous, binary, categorical). For binary or categorical variables, specify what the values mean.
3. Write a natural language causal question the dataset could answer. The question should:
 - Relate implicitly to the treatment and outcome
 - Avoid any statistical or causal terminology
 - Avoid naming variables directly
4. Write a 1–2 sentence summary capturing the dataset's overall intent and contents.

Return your output as a JSON object with the following keys:

- "variable_labels": {"X1": "education_years", ...}
- "description": "<paragraph>"
- "question": "<causal question>"

- "summary": <summary>
- "domain": "<domain>"

Return only a valid JSON object. Do not include any markdown, explanations, or extra text.

Notes on Placeholders

`summary` provides a description of the dataset. It specifies which symbols correspond to the treatment, outcome, continuous covariates, and binary covariates. It also adds method-specific details (e.g., IV instrument, RDD cutoff, or DiD setup) and includes a statistical summary of the variables as given by `df.describe()`.

`{history}` contains a record of previously generated dataset contexts. This is used to prevent duplication and ensure variety across generated scenarios.

`{domain_guides}` provides domain-specific guidance, such as reminding the model that education data often includes student performance and school-level features, or that healthcare data often covers treatments and recovery outcomes.

We verify that the output of the LLM does not explicitly describe the estimand or specify a causal inference method to be used. As an illustration, the following example shows a context and query generated by the LLM for an RCT dataset:

Example of a Query + Context for Synthetic Data

```
{
  "query": "Does providing housing subsidies improve the stability
of housing situations?",

  "dataset_description": "This dataset was compiled from a
Randomized Control Trial conducted by the Department of Housing
and Urban Development (HUD) of the United States. The goal was
to investigate the impact of a new housing subsidy policy on
recipients' housing stability. Variables include the age of the
recipient ('recipient_age'), their monthly income
('monthly_income'), whether they own a home ('is_homeowner',
binary: 1 for homeowners, 0 for non-homeowners), whether they
have dependents ('has_dependents', binary: 1 for yes, 0 for no),
whether they reside in a rural area ('lives_in_rural_area',
binary: 1 for rural, 0 for urban), whether they received the
housing subsidy ('received_subsidy', binary: 1 for yes, 0 for
no), and their self-reported housing stability
('housing_stability')."
}
```

D. Perturbation Analysis Prompts

We use the prompts below for applying perturbations and source identification.

Prompt for Applying Perturbations

Apply perturbations to the above prompt by changing the story completely and the variable names. Try to make sure the LLM can't recall which paper this came from. Output should be in this format: "Description: <new description> Natural Language Query: <new natural language query>. Do not change the name of the csv file."

Prompt for Source Identification

Which research paper or textbook does this research question and description come from? Think carefully and try recalling which paper or textbook it is from. Answer with 'yes' or 'no' if you know it or not, and if yes, write the name of the paper in " quotes -> e.g yes, The name of the paper is "paper name". Start with a 'yes' or 'no'!

E. Annotation Details

For each article we curate the following information:

- **Paper Name:** Name of the study
- **Description:** The description about the dataset that includes the collection process, purpose, and brief explanation about the variable names
- **Query:** Causal question associated with the dataset
- **Answer:** Causal effect derived in the paper
- **Standard Error:** Standard error associated with the causal effect estimate
- **Significant:** Binary variable indicating if the effect is statistically significant
- **Method:** The causal inference method
- **Treatment:** The name of the treatment variable in the dataset
- **Outcome:** The name of the outcome variable in the dataset
- **Control Covariates:** The control variables / confounders used in the estimation model
- **Interaction Variable:** The name of the variable that interacts with the treatment. This is used for measuring heterogeneous treatment effects
- **Instrument:** The variable used as an instrument. If instrumental variable is not used, this is set to null
- **Running Variable:** The running variable for Regression Discontinuity Design (RDD). If RDD is not used, we set this to null
- **Temporal Variable:** The variable denoting the timing of treatments. This is used for difference-in-differences
- **State Variable:** The variable denoting the different participating entities. This is used for two way fixed effects versions of difference in difference
- **Multi-RCT Treatment Variable:** The treatment type of interest. This is used in RCTs with multiple treatments
- **Data File:** The name of the csv file containing the data
- **Reference:** Reference to the original paper, where the result is found
- **Publication Year:** The year the original study was published
- **Domain:** The domain of the original study

F. Results Table

Below, we present the full table reporting all metrics across individual models and prompting strategies.

F.1. Method Selection Accuracy and Mean Relative Error

Method Selection Accuracy (MSA): Percentage of queries where the selected causal inference method $\hat{m}_i$ matches the reference method (m_i):

$$\text{MSA} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{m}_i = m_i] \times 100\%. \quad (4)$$

Mean Relative Error (MRE) is the average relative error between predicted causal effects ($\hat{\tau}_i$) and reference values (τ_i). Mathematically,

$$\text{MRE} = \frac{1}{N} \sum_{i=1}^N \min \left(\frac{|\hat{\tau}_i - \tau_i|}{|\tau_i|}, 1 \right) \times 100\% \quad (5)$$

To reduce the impact of outliers, relative error is capped at 100% per query.

In the above formulae, N is the number of queries for which the relevant quantities were successfully obtained.

Table 6. Comparison of method selection accuracy ($\uparrow$) and mean relative error ($\uparrow$) across datasets, models, and prompting strategies. For each model, **dark green** indicates the highest method selection accuracy and **dark red** indicates the lowest relative error for a given model across the 4 prompting approaches.

| Dataset | Model | Method Sel. Accuracy ($\uparrow$) | | | | Mean Rel. Error ($\downarrow$) | | | |
|-----------|-----------------------|-------------------------------------|--------------|--------------|--------------|----------------------------------|--------------|--------------|--------------|
| | | Direct | CoT | PoT | ReAct | Direct | CoT | PoT | ReAct |
| Real | Gemini-2.5-Flash-Lite | 42.86 | 51.63 | 51.91 | 54.34 | 67.64 | 66.08 | 71.82 | 72.09 |
| | Grok-4-Fast | 74.05 | 73.37 | 65.95 | 67.21 | 59.02 | 58.44 | 58.36 | 66.77 |
| | GPT-5-mini | 69.40 | 70.95 | 68.31 | 70.32 | 59.05 | 55.78 | 59.50 | 53.56 |
| | GPT-4o-mini | 32.58 | 36.93 | 36.52 | 35.43 | 76.01 | 75.44 | 76.17 | 68.02 |
| | GPT-4o | 56.52 | 59.34 | 56.14 | 59.41 | 67.09 | 67.44 | 74.20 | 66.58 |
| | OpenAI-o3 | 70.83 | 77.17 | 68.33 | 72.60 | 52.56 | 48.96 | 63.10 | 66.48 |
| | Qwen3-Next-80B-Inst | 56.76 | 62.70 | 57.07 | 55.80 | 61.57 | 67.52 | 70.65 | 68.64 |
| QRData-CI | Gemini-2.5-Flash-Lite | 61.54 | 76.32 | 71.79 | 82.05 | 40.43 | 35.82 | 38.80 | 49.23 |
| | Grok-4-Fast | 66.67 | 66.67 | 66.67 | 63.16 | 29.03 | 23.79 | 26.31 | 25.60 |
| | GPT-5-mini | 69.23 | 75.68 | 71.79 | 61.76 | 41.14 | 42.02 | 47.34 | 32.91 |
| | GPT-4o-mini | 51.28 | 62.16 | 60.53 | 54.05 | 36.51 | 31.21 | 37.43 | 26.29 |
| | GPT-4o | 61.54 | 66.67 | 66.67 | 61.54 | 43.72 | 42.33 | 26.76 | 33.58 |
| | OpenAI-o3 | 66.67 | 60.71 | 66.67 | 72.73 | 30.10 | 37.93 | 44.56 | 63.48 |
| | Qwen3-Next-80B-Inst | 74.36 | 76.92 | 84.62 | 74.36 | 35.55 | 43.74 | 42.25 | 38.60 |
| Synthetic | Gemini-2.5-Flash-Lite | 79.43 | 84.17 | 80.42 | 62.86 | 45.25 | 34.05 | 39.27 | 48.34 |
| | Grok-4-Fast | 76.81 | 76.76 | 69.72 | 71.74 | 15.86 | 13.08 | 23.64 | 32.14 |
| | GPT-5-mini | 87.77 | 91.43 | 93.48 | 84.68 | 7.91 | 7.93 | 14.33 | 9.40 |
| | GPT-4o-mini | 15.38 | 24.48 | 27.34 | 23.78 | 21.78 | 26.41 | 30.74 | 25.29 |
| | GPT-4o | 70.63 | 83.10 | 80.14 | 65.49 | 28.82 | 28.03 | 25.46 | 21.95 |
| | OpenAI-o3 | 86.47 | 91.35 | 79.58 | 80.00 | 8.43 | 43.97 | 17.20 | 71.57 |
| | Qwen3-Next-80B-Inst | 72.86 | 75.00 | 80.14 | 65.44 | 22.89 | 33.18 | 34.42 | 34.82 |

F.2. Looking at the coding capabilities of the models

The table below presents the completion rates of the models. It measures, on average, how often each model successfully generates executable code within three attempts.

Table 7. Completion rates (%) by dataset, prompt type, and model.

| Dataset | Prompt | Gemini-2.5-Flash-Lite | GPT-4o | GPT-4o-mini | GPT-5-mini | Grok-4-Fast | OpenAI-o3 | Qwen3-80B |
|-----------|--------|-----------------------|--------|-------------|------------|-------------|-----------|-----------|
| Real | Direct | 91.9 | 88.6 | 87.6 | 91.9 | 92.4 | 85.9 | 95.1 |
| | CoT | 93.5 | 93.5 | 87.6 | 84.3 | 93.5 | 37.3 | 95.7 |
| | PoT | 82.7 | 67.0 | 61.1 | 77.8 | 86.5 | 80.5 | 94.1 |
| | ReAct | 87.0 | 69.2 | 48.1 | 71.9 | 95.7 | 22.2 | 91.4 |
| QRData-CI | Direct | 97.4 | 100.0 | 89.7 | 100.0 | 100.0 | 100.0 | 100.0 |
| | CoT | 97.4 | 97.4 | 89.7 | 94.9 | 97.4 | 64.1 | 100.0 |
| | PoT | 100.0 | 92.3 | 76.9 | 87.2 | 100.0 | 100.0 | 100.0 |
| | ReAct | 100.0 | 92.3 | 84.6 | 87.2 | 94.9 | 15.4 | 97.4 |
| Synthetic | Direct | 93.7 | 95.1 | 98.6 | 88.8 | 83.9 | 89.5 | 92.3 |
| | CoT | 90.2 | 96.5 | 100.0 | 85.3 | 89.5 | 50.3 | 95.8 |
| | PoT | 92.3 | 88.1 | 93.7 | 87.4 | 92.3 | 94.4 | 99.3 |
| | ReAct | 97.2 | 95.8 | 100.0 | 77.6 | 95.1 | 9.8 | 93.7 |

F.3. Breaking down relative errors by method selection correctness

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Table 8. Impact of method selection on causal effect estimation error. Mean relative errors are averaged across all prompting strategies (Direct, CoT, PoT, ReAct). Percentage (%) indicate the proportion of examples with correct versus incorrect method selection. Diff. represents the difference in mean relative errors between correctly and incorrectly selected methods

| Model | Method Selection | Real | | | Synthetic | | | QRData | | |
|----------------|------------------|-------|------|-------|-----------|------|-------|--------|------|-------|
| | | Error | % | Diff. | Error | % | Diff. | Error | % | Diff. |
| Gemini-2.5-FL | Correct Method | 61.6 | 49.6 | +15.7 | 40.5 | 76.4 | +7.3 | 39.4 | 72.7 | +5.3 |
| | Incorrect Method | 77.3 | 50.4 | | 47.8 | 23.6 | | 44.7 | 27.3 | |
| Grok-4-Fast | Correct Method | 52.9 | 69.5 | +25.6 | 19.9 | 73.3 | +3.2 | 22.5 | 65.6 | +10.4 |
| | Incorrect Method | 78.5 | 30.5 | | 23.1 | 26.7 | | 32.9 | 34.4 | |
| GPT-5-mini | Correct Method | 47.9 | 71.5 | +31.6 | 7.6 | 90.5 | +23.6 | 51.5 | 70.1 | −36.4 |
| | Incorrect Method | 79.5 | 28.5 | | 31.2 | 9.5 | | 15.2 | 29.9 | |
| GPT-4o-mini | Correct Method | 67.5 | 35.1 | +9.7 | 11.0 | 23.0 | +19.6 | 26.3 | 58.2 | +15.0 |
| | Incorrect Method | 77.2 | 64.9 | | 30.5 | 77.0 | | 41.3 | 41.8 | |
| GPT-4o | Correct Method | 62.4 | 57.4 | +15.5 | 22.6 | 74.0 | +12.0 | 39.5 | 63.8 | −8.8 |
| | Incorrect Method | 77.8 | 42.6 | | 34.6 | 26.0 | | 30.7 | 36.2 | |
| OpenAI-o3 | Correct Method | 49.9 | 71.0 | +29.5 | 31.5 | 85.7 | +26.1 | 52.2 | 64.2 | −23.3 |
| | Incorrect Method | 79.4 | 29.0 | | 57.5 | 14.3 | | 28.9 | 35.8 | |
| Qw3-Next-80B-I | Correct Method | 60.2 | 57.9 | +16.1 | 32.5 | 73.9 | −5.6 | 33.2 | 77.4 | +30.9 |
| | Incorrect Method | 76.3 | 42.1 | | 26.9 | 26.1 | | 64.1 | 22.6 | |

F.4. Variable selection accuracy

Table 9. Treatment and outcome variable overlap scores

| Dataset | Model | Treatment Variable | | | | Outcome Variable | | | |
|-----------|-----------------------------|--------------------|--------|--------|--------|------------------|--------|--------|--------|
| | | Basic | CoT | PoT | ReAct | Basic | CoT | PoT | ReAct |
| Real | gemini-2.5-flash-lite | 76.32 | 76.32 | 73.68 | 80.56 | 97.14 | 97.06 | 91.43 | 97.06 |
| | gpt-4o-mini | 86.11 | 84.21 | 84.21 | 81.08 | 94.12 | 97.06 | 91.43 | 87.88 |
| | gpt-4 | 71.05 | 81.58 | 76.92 | 81.08 | 94.29 | 94.29 | 94.29 | 97.14 |
| | gpt-5-mini | 76.92 | 81.58 | 84.21 | 71.88 | 94.29 | 94.12 | 94.29 | 96.43 |
| | grok-4-fast | 76.32 | 78.38 | 84.21 | 78.95 | 97.14 | 97.06 | 97.14 | 94.12 |
| | o3 | 81.58 | 78.95 | 78.95 | 92.31 | 97.06 | 100.00 | 94.12 | 87.50 |
| | qwen3-next-80b-a3b-instruct | 73.68 | 78.95 | 76.32 | 66.67 | 94.29 | 94.29 | 97.14 | 93.94 |
| Synthetic | gemini-2.5-flash-lite | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| | gpt-4o-mini | 100.00 | 100.00 | 100.00 | 97.44 | 100.00 | 100.00 | 100.00 | 97.44 |
| | gpt-4 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| | gpt-5-mini | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 97.14 |
| | grok-4-fast | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| | o3 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| | qwen3-next-80b-a3b-instruct | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 |
| QRData | gemini-2.5-flash-lite | 85.71 | 82.35 | 91.43 | 82.86 | 92.31 | 92.11 | 89.74 | 89.74 |
| | gpt-4o-mini | 82.86 | 81.82 | 85.29 | 82.35 | 89.74 | 91.89 | 86.84 | 91.89 |
| | gpt-4 | 82.86 | 82.86 | 82.86 | 82.86 | 92.31 | 92.31 | 87.18 | 92.31 |
| | gpt-5-mini | 82.86 | 81.82 | 88.57 | 83.87 | 92.31 | 91.89 | 92.31 | 91.18 |
| | grok-4-fast | 85.71 | 82.86 | 88.57 | 84.85 | 92.31 | 92.31 | 92.31 | 91.89 |
| | o3 | 82.86 | 85.71 | 88.57 | 64.71 | 87.18 | 87.10 | 87.18 | 77.78 |
| | qwen3-next-80b-a3b-instruct | 82.86 | 82.86 | 82.86 | 80.00 | 92.31 | 92.31 | 92.31 | 89.74 |

F.5. Control variable overlap and code generation attempts

Code generation attempts: Number of attempts (capped at 3) it takes for the model to write error-free code.

Table 10. Control variable overlap scores and average number of code generation attempts

| Dataset | Model | Control Variable Overlap | | | | Code Generation Attempts | | | |
|-----------|-----------------------------|--------------------------|--------|-------|-------|--------------------------|------|------|-------|
| | | Basic | CoT | PoT | ReAct | Basic | CoT | PoT | ReAct |
| Real | gemini-2.5-flash-lite | 60.59 | 65.95 | 71.30 | 48.75 | 1.68 | 2.01 | 2.35 | 1.93 |
| | gpt-4o-mini | 63.12 | 69.43 | 68.79 | 51.11 | 2.56 | 2.57 | 2.27 | 1.08 |
| | gpt-4 | 71.92 | 61.64 | 53.29 | 36.19 | 2.31 | 2.21 | 2.26 | 1.38 |
| | gpt-5-mini | 61.20 | 64.46 | 69.82 | 74.67 | 2.56 | 2.56 | 2.56 | 1.98 |
| | grok-4-fast | 53.98 | 45.29 | 47.51 | 44.01 | 2.49 | 2.46 | 2.60 | 1.42 |
| | o3 | 57.61 | 56.74 | 69.40 | 45.22 | 2.69 | 2.62 | 2.56 | 2.85 |
| | qwen3-next-80b-a3b-instruct | 62.22 | 73.48 | 78.16 | 49.53 | 2.23 | 2.03 | 2.19 | 1.19 |
| Synthetic | gemini-2.5-flash-lite | 89.23 | 100.00 | 96.19 | 81.11 | 1.58 | 2.12 | 2.07 | 2.12 |
| | gpt-4o-mini | 100.00 | 99.57 | 98.72 | 98.80 | 2.94 | 2.97 | 2.92 | 1.06 |
| | gpt-4 | 92.31 | 82.05 | 96.92 | 73.85 | 2.21 | 2.22 | 2.54 | 1.31 |
| | gpt-5-mini | 89.23 | 95.96 | 99.49 | 99.49 | 2.78 | 2.84 | 2.74 | 2.29 |
| | grok-4-fast | 58.97 | 76.92 | 89.74 | 54.74 | 2.43 | 2.41 | 2.54 | 1.50 |
| | o3 | 89.23 | 89.23 | 96.92 | 92.86 | 2.54 | 2.67 | 2.66 | 2.79 |
| | qwen3-next-80b-a3b-instruct | 96.92 | 93.57 | 99.49 | 83.74 | 2.28 | 2.18 | 2.35 | 1.60 |
| QRData | gemini-2.5-flash-lite | 95.14 | 96.47 | 94.42 | 81.25 | 1.87 | 1.79 | 2.23 | 1.69 |
| | gpt-4o-mini | 96.33 | 95.58 | 96.38 | 94.31 | 2.80 | 2.77 | 2.57 | 1.18 |
| | gpt-4 | 96.91 | 96.80 | 96.91 | 90.38 | 2.67 | 2.66 | 2.83 | 1.50 |
| | gpt-5-mini | 97.01 | 96.30 | 96.67 | 96.91 | 2.90 | 2.84 | 2.76 | 1.94 |
| | grok-4-fast | 81.46 | 74.53 | 90.95 | 74.93 | 2.77 | 2.84 | 2.82 | 1.68 |
| | o3 | 97.11 | 96.93 | 96.80 | 70.67 | 2.92 | 2.96 | 2.87 | 3.00 |
| | qwen3-next-80b-a3b-instruct | 95.61 | 94.61 | 96.37 | 94.61 | 2.46 | 2.44 | 2.67 | 1.42 |

G. Pass@3 Results Table

We performed pass@3 evaluation using 2 models on a representative subset of 100 queries in Real and Synthetic, and the full 39 for Textbook.

Table 11. Pass@1 vs. Pass@3 per prompting strategy for GPT-4o and GPT-4o-mini. Bold values indicate the largest Pass@3 gain within each row.

| Dataset | Model | Method Sel. Accuracy Pass@1 / Pass@3 (↑) | | | | Mean Rel. Error Pass@1 / Pass@3 (↓) | | | |
|-----------|-------------|--|-------------|-------------|-------------|-------------------------------------|-------------|-------------|-------------|
| | | Direct | CoT | PoT | ReAct | Direct | CoT | PoT | ReAct |
| Real | GPT-4o | 58.6 / 62.6 | 57.6 / 68.7 | 55.6 / 66.7 | 46.5 / 61.6 | 70.8 / 54.8 | 69.3 / 57.1 | 70.9 / 57.7 | 55.0 / 52.8 |
| | GPT-4o-mini | 29.3 / 35.4 | 26.3 / 29.3 | 31.3 / 41.4 | 32.3 / 36.4 | 71.4 / 59.6 | 69.7 / 61.9 | 66.3 / 59.5 | 68.2 / 63.0 |
| Synthetic | GPT-4o | 90.0 / 96.0 | 92.0 / 97.0 | 82.0 / 93.0 | 62.0 / 70.0 | 22.5 / 15.3 | 22.6 / 16.5 | 20.7 / 11.5 | 10.0 / 11.9 |
| | GPT-4o-mini | 17.0 / 17.0 | 25.0 / 26.0 | 30.0 / 37.0 | 27.0 / 28.0 | 16.6 / 15.8 | 21.3 / 15.8 | 21.7 / 13.6 | 22.3 / 16.7 |
| QRData-CI | GPT-4o | 53.8 / 61.5 | 69.2 / 71.8 | 61.5 / 76.9 | 53.8 / 69.2 | 53.5 / 42.9 | 41.3 / 32.9 | 55.3 / 33.3 | 43.7 / 28.0 |
| | GPT-4o-mini | 51.3 / 53.8 | 56.4 / 59.0 | 53.8 / 79.5 | 59.0 / 59.0 | 37.0 / 27.9 | 42.6 / 32.1 | 35.1 / 24.6 | 29.7 / 29.5 |

H. Contamination analysis

Using existing studies and datasets available online raises concerns about contamination. To address this, we conducted a perturbation study (Zhang et al., 2025; 2026) on 39 queries from both the “Real” and “QRData” data, respectively. The perturbations leverage GPT-4o to modify the narratives of the dataset descriptions, the variables, and, accordingly, the queries and the column labels of the CSV files. Then, we re-ran the full ReAct pipeline on the perturbed datasets. In the process, we manually reviewed the resulting samples to ensure that the narratives differed from the originals and that there were no inconsistencies. Additionally, we conducted a direct recall test, querying the LLM to determine whether it could recall the source paper both for the original and perturbed samples. The prompts used for both the perturbations and the source-identification experiments are provided in Section D.

Table 12 shows the results for 39 queries in the Real and QRData datasets. Across both datasets, accuracy and relative error remain fairly high, while source identification drops to near zero. This indicates that the models are not relying on memorization. We also observed that, although the source identification rate was slightly higher in the Real dataset, many of these cases involved hallucinated names rather than the actual sources. The combination of stable performance under perturbation and low direct recall rates shows that memorization is not influencing our evaluations.

Table 12. Performance of GPT-4o and GPT-4o-mini on 39 queries of Real, Perturbed Real, Original QRData, and Perturbed QRData datasets. Accuracy (%) and Relative Error are reported, with lower values for Source Identification indicating better performance.

| Dataset Version | Model | Method Selection Accuracy (%) | Relative Error | Source Identification (lower is better) |
|------------------|-------------|-------------------------------|----------------|---|
| Real | GPT-4o-mini | 24.32 | 59.64 | 11/39 |
| Perturbed Real | GPT-4o-mini | 22.22 | 49.53 | 1/39 |
| Real | GPT-4o | 67.57 | 50.56 | 12/39 |
| Perturbed Real | GPT-4o | 53.12 | 62.69 | 0/39 |
| Original QRData | GPT-4o-mini | 56.76 | 26.67 | 19/39 |
| Perturbed QRData | GPT-4o-mini | 56.76 | 33.84 | 0/39 |
| Original QRData | GPT-4o | 50.00 | 42.45 | 9/39 |
| Perturbed QRData | GPT-4o | 52.94 | 33.97 | 0/39 |

I. Notes on Causal Inference Methods

This section offers a brief overview of the causal inference approaches we examine. For comprehensive theoretical foundations and detailed methodological discussions, we direct readers to textbooks on causal inference (Cunningham, 2021; Imbens & Rubin, 2015; Hernan & Robins, 2025; Peters et al., 2017).

I.1. Randomized Control Trials

RCTs are the gold standard for causal inference. This is because the ignorability assumption, which states that treatment assignment is independent of the potential outcomes, is satisfied by default.

$$Y(0), Y(1) \perp T \quad (6)$$

Identification Assumption The key assumption is ignorability (6).

Assumption Check Whether or not data comes from an RCT should be specified in the data description. Ignorability is enforced by design. On average, the characteristics of units in the treatment and control units are the same. SUTVA is also enforced by design in most cases.

Estimand The causal estimand of interest is Average Treatment Effect (ATE).

Causal Estimation The most straightforward way to estimate causal effect is **Difference in means**. As the name states, we simply find the difference in the average outcomes for treatment and control groups. Mathematically,

$$\hat{\tau} = \sum_{i \in \text{Treatment}} \frac{1}{n_1} Y_i - \sum_{i \in \text{Control}} \frac{1}{n_0} Y_i \quad (7)$$

where n_1 and n_0 are the total number of units in treatment and control groups respectively.

In practice, we often compute $\hat{\tau}$ by regressing the outcome (Y) on treatment (T).

In some cases, the data may also contain pre-treatment covariates. These are variables measured before the experiment and are unaffected by the treatment. We often include them in our estimation to improve the precision of the causal effect measure, i.e., minimize the standard error. In such cases, the causal effect model is

$$Y = \alpha + \tau T + X\beta + \epsilon \text{ where } \epsilon \text{ is an error term uncorrelated with } T \text{ and } X \quad (8)$$

I.2. IPW

Inverse Probability Weighting (IPW) is one of the methods for estimating causal effects from observational datasets. The key assumption underlying IPW is conditional ignorability. This states that the potential outcomes are independent of treatment assignment conditioned on confounding variables. Confounding variables are those that affect both treatment and outcome. Mathematically,

$$Y(0), Y(1) \perp T | X \quad (9)$$

Propensity Score Propensity score, $e(X) \in [0, 1]$, gives a measure of how likely a unit is to be treated. To estimate propensity scores, we can fit logit or probit models on the confounders X. Upon computing the propensity scores, we can directly compute IPW estimates. Note that when fitting the propensity scores, you should fit a single model for the whole data.

Estimand Average Treatment Effect (ATE), Average Treatment Effect on the Treated (ATT), Average Treatment Effect on the Control (ATC). The right estimand depends from problem to problem. The most popular estimand is ATT, then ATE and ATC.

Assumption The key assumption is conditional ignorability.

Causal Estimation The measures of causal effects are

$$\hat{\tau}_{ATE} = \frac{\sum_{i:T_i=1} \frac{Y_i}{e(X_i)}}{\sum_{i:T_i=1} \frac{1}{e(X_i)}} - \frac{\sum_{i:T_i=0} \frac{Y_i}{1-e(X_i)}}{\sum_{i:T_i=0} \frac{1}{1-e(X_i)}} \quad (10)$$

$$\hat{\tau}_{ATT} = \frac{1}{n_1} \sum_{i:T_i=1} Y_i - \frac{\sum_{i:T_i=0} \frac{e(X_i)}{1-e(X_i)} Y_i}{\sum_{i:T_i=0} \frac{e(X_i)}{1-e(X_i)}} \quad (11)$$

$$\hat{\tau}_{ATC} = \frac{\sum_{i:T_i=1} \frac{1-e(X_i)}{e(X_i)} Y_i}{\sum_{i:T_i=1} \frac{1-e(X_i)}{e(X_i)}} - \frac{1}{n_0} \sum_{i:T_i=0} Y_i \quad (12)$$

Assumption Check We need to satisfy the conditional ignorability assumption. This is an untestable assumption. Experts usually use their domain knowledge to select confounding variables and justify their selection.

Another method to check is to assess the distribution of covariates in the treated and control groups. A popular method for assessing the distribution is SMD (Standardized Mean Difference). For each covariate x , we compute the standardized mean difference between treatment and control groups as:

$$SMD_x = \frac{\mu_x^{\text{treatment}} - \mu_x^{\text{control}}}{\sqrt{(\sigma_x^{\text{treatment}})^2 + (\sigma_x^{\text{control}})^2} / 2}$$

where μ_x and σ_x are the mean and standard deviation of covariate x in each group. If ignorability is approximately satisfied, the standardized mean difference should be close to zero for each confounder.

Likewise, we can also assess the propensity scores for treated and control groups. Ideally, we want the distribution of the propensity scores to be similar. If there are many confounders, we often compute the SMD for propensity scores.

Another method to assess the distribution of confounders is through visual inspection.

I.3. Matching

As stated above, propensity score based estimators are highly unstable for real world problems. To improve stability, we often use matching. As the name suggests, we match each unit with its nearest neighbor. The causal effect for that particular unit is

$$\tau_i = Y_i - Y_{m_i} \quad \text{where } m_i \text{ is the unit matched to } i$$

You can think of matching as an equivalent of the nearest neighbor method in machine learning. Matching requires computing similarities. One common way to perform matching is to match units with similar propensity scores.

Type of Matching Just like in the nearest neighbor method, we have k-matching, i.e., for each unit, we select the K nearest neighbors, and then compute the causal effect as

$$\tau_i = Y_i - \frac{1}{K} \sum_{k=1}^K Y_k \quad (13)$$

Similarly, we can have matching with replacement or without replacement. Matching with replacement is more common in practice.

Causal Estimation You can think of matching as a preprocessing step. We compute the causal effect using the matched units. The nature of matching varies between estimands.

- **ATE** Each unit in control is matched to a unit in the treatment group and vice versa. To compute the causal effect,

$$\hat{\tau}_{ATE} = \frac{1}{N} \sum_{i=1}^N (Y_i - Y_{m_i}) \quad (14)$$

Notice that we do not compute the means for treatment and control separately.

- **ATT** We only match units in the treatment group, i.e., for each unit in the treatment group, we select k nearest neighbors in the control group. The causal effect is then computed as:

$$\hat{\tau}_{ATT} = \frac{1}{n_1} \sum_{i \in \text{Treatment}} \left(Y_i - \frac{1}{K} \sum_{k=1}^K Y_{m_{i,k}} \right) \quad (15)$$

where $m_{i,k}$ denotes the k-th matched control unit for treated unit i.

I.3.1. MATCHING VS IPW

IPW is fast and relatively easier to implement. However, IPW is highly unstable when the overlap assumption is violated. Hence, in practice we often use matching. If the propensity scores are well balanced across both the control and treated units, the causal effect from matching and IPW should be similar to one another. In such cases, the estimates from IPW would be fairly reliable. However, if the balance is poor, we should use matching. Thus, you can think of matching as a preprocessing step that improves the comparability of treatment and control groups.

I.4. Difference in Differences (DiD)

Difference in Differences is a quasi-experimental method for computing causal effects by addressing time-invariant confounding. For instance, suppose New Jersey raised its minimum wage, but its neighbor Pennsylvania did not. We are interested in the impact of the wage policy on employment rates across two time periods. Over time, several things could have changed that could impact employment rates, for example, tax policies. In such cases, we can use the trends in Pennsylvania to remove the effects from other time-varying factors and compute the impact of the minimum wage policy.

For DiD to be considered, we must have panel data, i.e., observations across multiple time periods (at least two). Moreover, the time-related information must not be a covariate. It must correspond to the timing of the treatments.

Identification Assumptions Two key assumptions underlying DiD are:

- **Parallel Trends Assumption** In the absence of treatment, the outcome trends in treatment and control groups would have evolved similarly, i.e., if there was no treatment, $E[Y_{1t} - Y_{1t-1} | \text{Treatment}] = E[Y_{0t} - Y_{0t-1} | \text{Control}]$. This states that the change in outcome in the two groups would be the same in the counterfactual scenario.
- **No Anticipatory Effects** This states that the treatment effect applies only after implementation, meaning units do not change their behavior in anticipation of future treatment.

Assumption Check First, we need to test if DiD is the right method. Just because the data has a time-related variable does not mean DiD is the right method. One should first identify what the treatment variable is, and then check if the time-related variable indicates the treatment timing.

The **no anticipatory effects** assumption is typically valid by design if the treatment timing is exogenous. The assumption of primary interest is the parallel trends assumption. We can test for this visually. For instance, if we have data on the outcomes for more than two periods before treatment, we can plot them and examine the slopes of the lines in the two groups. If the parallel trends assumption is valid, then the slopes should be similar, i.e., roughly parallel. In case we do not have enough pre-treatment data, say there are only two time periods (pre and post), then we can use domain knowledge to justify why parallel trends could be valid.

Estimand The estimand is the Average Treatment Effect on the Treated (ATT).

Causal Estimation There are two main scenarios under DiD:

- **Canonical DiD** This is the classical 2 period and 2 group setting. This means we have 2 time periods: pre and post treatment periods, and 2 groups: treatment and control. This applies in situations where the treatment was applied at a single point in time to a specific group. For causal effect estimation, we define two indicator variables:

1. **POST**: indicator variable that is 1 if the observation is made after treatment is applied, i.e., $POST_t = 1$ if $t \geq \text{treatment time}$
2. **TREAT**: indicator variable that is 1 if unit i is in the treatment group and 0 otherwise.

Then the causal model is:

$$Y_{i,t} = \alpha + \beta \cdot POST_t + \gamma \cdot TREAT_i + \delta \cdot POST_t \times TREAT_i + X_{i,t}\beta + \epsilon_{i,t} \quad (16)$$

The coefficient of interest is δ , which represents the DiD estimator. We can add a valid set of control variables $X_{i,t}$. Note that these must not be affected by the treatment (bad controls).

- **Two-Way Fixed Effects (TWFE)** This is a generalized version of DiD, where the treatment is staggered, i.e., there are multiple units receiving treatment at different periods. An example of this could be the adoption of unilateral divorce laws by US states in different years. This takes the following form:

$$Y_{i,t} = \alpha_i + \lambda_t + \delta \cdot D_{i,t} + X_{i,t}\beta + \epsilon_{i,t} \quad (17)$$

The key coefficient of interest is δ . $D_{i,t}$ is an indicator variable that is 1 if unit i has received treatment by time t . α_i captures unit-specific fixed effects and λ_t captures time-specific fixed effects.

I.5. Regression Discontinuity Design (RDD)

Regression Discontinuity Design is another quasi-experimental method. It is applicable in situations where treatment assignment is dictated by a threshold value. For instance, say we want to evaluate the impact of stimulus checks on total household spending. We could exploit the fact that stimulus checks are given to people whose annual income is less than 70k.

In RDD, the variable that determines treatment assignment is called the running variable (r_i). If the cutoff is r_0 , the treatment assignment is

$$T_i = \begin{cases} 1 & \text{if } r_i \geq r_0 \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

The causal effect is thus

$$\tau_{RDD} = \lim_{r \rightarrow r_0^+} E[Y | R = r] - \lim_{r \rightarrow r_0^-} E[Y | R = r]$$

This means we are computing the causal effect around the threshold value.

Assumption The identification assumption is that around the cutoff the potential outcomes are continuous. The outcomes change abruptly only when we transition from control to treatment at the cutoff.

Assumption check Usually, we perform visual inspection around the cutoff. This means fitting curves to the left and right of the cutoff based on the running variable, and looking for jumps between the curves at the threshold.

I.6. Instrumental Variables (IV)

This is another popular method for causal effect estimation. It is useful in cases with unobserved confounders. We use an instrument, which is a variable that affects the outcome only through the treatment and is independent of unobserved confounders.

IV applies to both randomized and non-randomized experiments.

- **Encouragement Design** This is the randomized case where treatment assignment is random but not all assignees accept their treatment, i.e., assignment is not the same as uptake. In such cases, we estimate the causal effect for the compliers (units who comply with their assignment). To check if this is an encouragement design, verify that the data come from a randomized experiment and that compliance information is available.
- **General IV** This describes quasi-experimental settings where we can find a variable that influences treatment but is not affected by unobserved confounders.

Estimand The estimand in Encouragement Design IV is called LATE (Local Average Treatment Effect) or CACE (Complier Average Causal Effect).

Assumption The assumptions underlying IV are:

- Independence of the instrument: the instrument is independent of potential outcomes (as if random).
- Exclusion restriction and relevance: the instrument affects the outcome only through the treatment (exclusion), and it is correlated with the treatment (relevance).
- Monotonicity: the instrument moves treatment in the same direction for all units (no defiers). For instance, being selected for the draft via lottery should not cause some otherwise willing participants to avoid service while motivating others to serve.

Assumption test Monotonicity and independence are usually justified by design and domain knowledge. In randomized encouragement designs, this is straightforward; otherwise, it requires a domain-based justification. To test relevance (instrument correlated with treatment), we can compute the F-statistic. The most important assumption, the exclusion restriction, is untestable and relies on substantive knowledge.

Causal Estimation There are two ways to compute causal effects.

- Non-parametric: most suitable for encouragement designs.

$$\hat{\tau}_{CACE} = \frac{\frac{\sum_{i:Z_i=1} Y_i}{\sum_i Z_i} - \frac{\sum_{i:Z_i=0} Y_i}{\sum_i (1 - Z_i)}}{\frac{\sum_{i:Z_i=1} D_i}{\sum_i Z_i} - \frac{\sum_{i:Z_i=0} D_i}{\sum_i (1 - Z_i)}} \quad (19)$$

- Parametric: The classic econometric approach (2 stage least squares / 2SLS), where we first regress the treatment on the instrument, then regress the outcome on the predicted value of treatment from the first stage. Intuitively, we use the part of treatment induced by the instrument to estimate the causal effect.

1.7. Backdoor and Frontdoor Adjustment

The above methods focus on the potential outcomes framework. Another approach to causal inference based on Pearl’s principles (Pearl, 2009) uses causal graphs. Causal graphs are DAGs that show how variables are causally related to one another. The general structure is *Cause* → *Effect*.

For graph-based methods, the first task is to construct a causal graph, and domain knowledge comes into play here. We also draw attention to the field of causal discovery, which is mainly concerned with learning causal graphs in a principled manner from data. However, it is important to distinguish causal discovery from causal effect estimation. While causal discovery focuses on identifying causal relationships and graph structure, causal effect estimation assumes the causal structure is known and focuses on quantifying the magnitude of causal effects.

Estimand Backdoor-based methods can estimate ATE, ATT, or ATC. Meanwhile, frontdoor computes the ATE.

Assumption Testing There is no definitive external test here. Backdoor and frontdoor adjustments rely on assumptions encoded by the DAG, many of which are untestable from observational data. The validity of the results hinges on the correctness of the graph; domain knowledge is typically used to justify it.

Link to previous methods Matching, IPW, and backdoor adjustment are interconnected approaches. Backdoor adjustment provides a systematic framework for choose the model variables from the graph aka the adjustment set. For instance, it allows us to select the confounders that we could use for matching, propensity score computation, etc. Using this adjustment set for effect estimation justifies how the conditional ignorability assumption is met.

Hence, the focus on backdoor criterion is on identification: i.e. what variables allows us to measure causal effects in a principled manner. For estimation, we then call methods like IPW or regression on the adjustment set. For a broader discussion on the connection between potential outcomes and Pearl’s framework, we refer the readers to Imbens (2020)

J. Data Sources

Here, we list papers we have used in our benchmark. We include citations for both the publication and the dataset.

| Publication Name | Publication Citation | Data Citation |
|---|-------------------------------|--|
| Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment | (Gerber et al., 2008b) | (Gerber et al., 2008a) |
| Propensity Score Matching for methods for non-experimental causal studies | (Dehejia & Wahba, 2002) | (Dehejia & Wahba, 1999; 2002; LaLonde, 1986) |
| Can immigrants counteract employer discrimination? A factorial field experiment reveals the immutability of ethnic hierarchies | (Vernby & Dancygier, 2019) | (Vernby, 2019) |
| Using geographic variation in college proximity to estimate the return to schooling | (Card, 1993) | (Huntington-Klein & Barrett, 2024) |
| Randomized experiments from non-random selection in U.S. House elections | (Lee, 2008) | (Olivares & Sarmiento-Barbieri, 2022) |
| The Long-run Effect of Abortion on Sexually Transmitted Infections | (Cornwell & Cunningham, 2013) | (Cunningham, 2021; Huntington-Klein & Barrett, 2024) |
| Black Politicians Are More Intrinsically Motivated to Advance Blacks’ Interests: A Field Experiment Manipulating Political Incentives | (Broockman, 2013) | (Huntington-Klein & Barrett, 2024) |
| Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Castle Doctrine | (Cheng & Hoekstra, 2012) | (Huntington-Klein & Barrett, 2024) |
| Government Transfers and Political Support | (Manacorda et al., 2011) | (Huntington-Klein & Barrett, 2024) |
| Don’t Take ‘No’ For An Answer: An Experiment With Actual Organ Donor Registrations | (Kessler & Roth, 2014) | (Huntington-Klein & Barrett, 2024) |
| The Demand for, and Impact of, Learning HIV Status | (Thornton, 2008) | (Huntington-Klein & Barrett, 2024) |
| Do Voters Affect or Elect Policies: Evidence from the U.S. House | (Lee et al., 2004) | (Huntington-Klein & Barrett, 2024; Cunningham, 2025; 2021) |
| The effects of rural electrification in India: An instrumental variable approach at the household level | (Thomas et al., 2020b) | (Thomas et al., 2020a) |
| Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania | (Card & Krueger, 1993) | (Shea, 2024) |
| Matching as Nonparametric Preprocessing for Reducing Model Dependence in Parametric Causal Inference | (Ho et al., 2007a) | (Ho et al., 2007b) |
| Markets: The Fulton Fish Market | (Graddy, 2006) | (Cunningham, 2025) |

| Publication Name | Publication Citation | Data Citation |
|--|------------------------------------|----------------------------|
| Punishment and Deterrence: Evidence from Drunk Driving | (Hansen, 2015) | (Cunningham, 2025) |
| The causal effect of economic sanctions on political stability: A two-stage difference-in-differences analysis | (Tan, 2025) | (Tan, 2024) |
| Public Trust and Collaborative Governance: An Instrumental Variable Approach | (Liu, 2024) | (Liu, 2022) |
| Does Compulsory School Attendance Affect Schooling and Earnings? | (Angrist & Krueger, 1990) | (Angrist & Krueger, 2009) |
| Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement | (Angrist & Lavy, 1997) | (Angrist & Lavy, 2009) |
| The China Syndrome: Local Labor Market Effects of Import Competition in the United States | (Autor et al., 2013) | (Autor et al., 2019) |
| Do Wall Street Landlords Undermine Renters' Welfare? | (Gurun et al., 2022b) | (Gurun et al., 2022a) |
| Privatization and Quality of Carceral Healthcare: A difference-in-differences analysis of jails in the United States, 2008-2019 | (Zoorob, 2024) | (Zoorob, 2022) |
| Estimating the impact of gubernatorial partisanship on policy settings and economic outcomes: A regression discontinuity approach | (Leigh, 2008) | (Leigh, 2013) |
| Do Congressional Candidates Have Reverse Coattails? Evidence from a Regression Discontinuity Design | (Broockman, 2017) | (Broockman, 2016a) |
| Early Medicaid Expansions and Drug Overdose Mortality in the USA: a Quasi-experimental Analysis | (Venkataramani & Chatterjee, 2018) | (Venkataramani, 2018) |
| Metrics Management and Bureaucratic Accountability: Evidence from Policing | (Eckhouse, 2022) | (Eckhouse, 2021) |
| Double-Shift Schooling and Student Success: Quasi-experimental Evidence from Europe | (Lusher & Yasenov, 2016) | (Yasenov, 2022) |
| Are Minimum Wages a Silent Killer? New Evidence on Drunk Driving Fatalities | (Sabia et al., 2019) | (Sabia et al., 2018) |
| Understanding the Impact of the 2018 Voter ID Pilots on Turnout at the London Local Elections: A Synthetic Difference-in-Difference Approach | (Barton, 2025a) | (Barton, 2025b) |
| The Minimum Legal Drinking Age and Morbidity in the US | (Carpenter & Dobkin, 2017) | (Carpenter & Dobkin, 2016) |
| How Partisan Is Local Law Enforcement? Evidence from Sheriff Cooperation with Immigration Authorities | (Thompson, 2020b) | (Thompson, 2020a) |

| Publication Name | Publication Citation | Data Citation |
|---|----------------------------------|----------------------------------|
| Immigration Policies and Access to the Justice System: The Effect of Enforcement Escalations on Undocumented Immigrants and Their Communities | (Dhingra et al., 2022) | (Dhingra et al., 2020) |
| The effect of number of siblings and birth order on educational attainment: Empirical Evidence from Chinese General Social Survey | (Xiong et al., 2020) | (Feng et al., 2020) |
| Do Televised Presidential Ads Increase Voter Turnout? Evidence from a Natural Experiment | (Krasno & Green, 2008b) | (Krasno & Green, 2008a) |
| Voting Made Safe and Easy: The Impact of e-voting on Citizen Perceptions | (Alvarez et al., 2013) | (Alvarez et al., 2015) |
| Can Learning Constituency Opinion Affect How Legislators Vote? Results from a Field Experiment | (Butler & Nickerson, 2011a) | (Butler & Nickerson, 2011b) |
| Doing Well by Doing Good: The Impact of Foreign Aid on Foreign Public Opinion | (Goldsmith et al., 2014) | (Goldsmith et al., 2018) |
| Were Newspapers More Interested in pro-Obama Letters to the Editor in 2008? Evidence from a Field Experiment | (Butler & Schofield, 2010a) | (Butler & Schofield, 2010b) |
| Foreign Direct Investors as Agents of Economic Transition: An Instrumental Variables Analysis | (Malesky, 2009) | (Malesky, 2015a) |
| Timing Is Everything? Primacy and Recency Effects in Voter Mobilization Campaigns | (Panagopoulos, 2011) | (Panagopoulos, 2010) |
| Computational and Robustness Reproducibility of "UN Peacekeeping and Democratization in Conflict-Affected Countries" | (Oswald & Walterskirchen, 2024b) | (Oswald & Walterskirchen, 2024a) |
| Incumbents Beware: The Impact of Offshoring on Elections | (Rickard, 2022) | (Rickard, 2021) |
| Does mislabeling COVID-19 elicit the perception of threat and reduce blame? | (Xu & Liu, 2021b) | (Xu & Liu, 2021a) |
| Undermining U.S. reputation: Chinese vaccines and aid and the alternative provision of public goods during COVID-19 | (Urdinez, 2024) | (Urdinez, 2023) |
| China's Foreign Aid Political Drivers: Lessons from a Novel Dataset of Mask Diplomacy in Latin America during the COVID-19 Pandemic | (Telias & Urdinez, 2022) | (Urdinez, 2021) |
| Regression Discontinuity Designs Using Covariates | (Calonico et al., 2019) | (Calonico et al., 2018) |
| The influence of waiting times and sociopolitical variables on public trust in healthcare: A cross-sectional study of the NHS in England | (Dorussen et al., 2024) | (Pickering, 2023) |

| Publication Name | Publication Citation | Data Citation |
|--|-------------------------------|-------------------------------|
| Europeanisation beyond the EU: Tobacco Advertisement Restrictions in Swiss Cantons | (Trein, 2017) | (Trein, 2016) |
| Does 'right to work' imperil the right to health? The effect of labour unions on workplace fatalities | (Zoorob, 2018b) | (Zoorob, 2018a) |
| The Effect of Property Assessment Reductions on Tax Delinquency and Tax Foreclosure | (Alfaro et al., 2025) | (Skidmore, 2025) |
| The Impact of Recentralization on Public Services: A Difference-in-Differences Analysis of the Abolition of Elected Councils in Vietnam | (Malesky et al., 2013) | (Malesky, 2015b) |
| Zika Epidemic and Birth Rates in Brazil | (Taddeo et al., 2022) | (Amorim, 2022) |
| The impact of intergenerational support on multidimensional poverty in old age | (Tan et al., 2023) | (Tan, 2023) |
| Internet-accessed sexually transmitted infection (e-STI) testing and results service: A randomised, single-blind, controlled trial | (Wilson et al., 2017) | (Wilson et al., 2017) |
| Estimating Incumbency Effects Using Regression Discontinuity Design | (Song, 2018) | (Song, 2019) |
| Experimental Evidence That Changing Beliefs About Mask Efficacy and Social Norms Increase Mask Wearing for COVID-19 Risk Reduction: Results From the United States and Italy | (Bokemper et al., 2021) | (Bokemper, 2021) |
| Vouchers for Private Schooling in Colombia: Evidence from a Randomized Natural Experiment | (Angrist et al., 2002) | (Angrist et al., 2009) |
| Instrumental Variables Estimates of the Effect of Subsidized Training on the Quantiles of Trainee Earnings | (Abadie et al., 1999) | (Abadie et al., 2009) |
| Field Experiments Testing the Impact of Radio Advertisements on Electoral Competition | (Panagopoulos & Green, 2008b) | (Panagopoulos & Green, 2008a) |
| Demand for in-app purchases in mobile apps—A difference-in-difference approach | (Enache et al., 2023) | (Friberg et al., 2023) |
| Comparing public support for nuclear and wind energy in Washington State | (Uji et al., 2023b) | (Uji et al., 2023a) |
| Does rising income inequality affect mortality rates in advanced economies? | (Rebeira et al., 2017b) | (Rebeira et al., 2017a) |
| Inference with 'Difference in Differences' with a Small Number of Policy Changes | (Conley & Taber, 2005) | (Taber & Conley, 2012) |
| Dismantling the "Jungle": migrant relocation and extreme voting in France | (Vertier et al., 2023) | (Gamalerio, 2022) |
| The corruption-income inequality trap: a study of Asian countries | (Dwiputri et al., 2018b) | (Dwiputri et al., 2018a) |

| Publication Name | Publication Citation | Data Citation |
|---|-------------------------------------|-------------------------------------|
| A Blame Shifting in Presidential Systems: Ministerial Terminations' Corrective Effect on Approval | (González-Bustamante, 2025b) | (González-Bustamante, 2025a) |
| Evaluating the Minority Candidate Penalty with a Regression Discontinuity Approach | (White et al., 2024) | (White, 2023) |
| The global costs of extreme weather that are attributable to climate change | (Newman & Noy, 2022) | (Noy, 2023) |
| Colored Perceptions: Racially Distinctive Names and Assessments of Skin Color | (Garcia & Abascal, 2016) | (Garcia & Abascal, 2018) |
| Social Capital, Racial Context, and Incarcerations in the American States | (Hawes, 2017c) | (Hawes, 2017b) |
| Give Us Your Tired, Your Poor and We Might Buy Them Dinner: Social Capital, Immigration, and Welfare Generosity in the American States | (Hawes & McCrea, 2018) | (Hawes, 2017a) |
| Impact of the Out-of-School Nutrition and Physical Activity (OSNAP) Group Randomized Controlled Trial on Children's Food, Beverage, and Calorie Consumption among Snacks Served | (Lee et al., 2018) | (Lee, 2019) |
| Partisan conflict over content moderation is more than disagreement about facts | (Appel et al., 2023a) | (Appel et al., 2023b) |
| A Call to Arms: How Rebel Groups Choose Their Recruitment Appeals | (Soules & Berlin, 2025) | (Soules, 2025) |
| The Effect of Microinsurance on Economic Activities: Evidence from a Randomized Field Experiment | (Cai et al., 2015) | (Cai et al., 2014) |
| Does Electing Women Reduce Corruption? A Regression Discontinuity Approach | (Pereira & Fernandez-Vazquez, 2023) | (Pereira & Fernandez-Vazquez, 2022) |
| Do Term Limits Restrain State Fiscal Policy? Approaches for Causal Inference in Assessing the Effects of Legislative Institut | (Keele et al., 2013b) | (Keele et al., 2013a) |
| Evaluating the Econometric Evaluations of Training Programs with Experimental Data | (LaLonde, 1986) | (Dehejia & Wahba, 1999; 2002) |
| Direct and Indirect Effects Based on Difference-in-Differences With an Application to Political Preferences Following the Vietnam Draft Lottery | (Deuchert et al., 2019) | (Huber, 2019) |
| Cities as Lobbyists | (Goldstein & You, 2017) | (You & Goldstein, 2017) |
| Do female politicians empower women to vote or run for office? A regression discontinuity approach | (Broockman, 2014) | (Broockman, 2016b) |
| Partisan Imbalance in Regression Discontinuity Studies Based on Electoral Thresholds | (Snyder et al., 2015) | (Snyder et al., 2013) |

| Publication Name | Publication Citation | Data Citation |
|---|---------------------------------|---------------------------------|
| The Causal Effect of Radical Right Success on Mainstream Parties' Policy Positions. A Regression Discontinuity Approach | (Abou-Chadi & Krause, 2020) | (Abou-Chadi & Krause, 2018) |
| Does earmarked funding affect the performance of international organisations? | (Reinsberg & Siauwijaya, 2024) | (Reinsberg & Siauwijaya, 2023) |
| Cabinets, Committees, and Careers: The Causal Effect of Committee Service | (Cirone & Van Coppenolle, 2018) | (Cirone & Van Coppenolle, 2017) |
| Improving Balance in Regression Discontinuity Design by Matching: Estimating the Effect of Academic Probation after the First Year of College | (Chi & Dow, 2014) | (Chi, 2014) |
| Deep IV: A Flexible Approach for Counterfactual Prediction | (Hartford et al., 2017) | (Hartford et al., 2017) |
| Privatization and Quality of Carceral Healthcare: A DiD Analysis of U.S. Jails (2008–2019) | (Zoorob, 2024) | (Zoorob, 2022) |
| All in the Family: Partisan Disagreement and Electoral Mobilization in Intimate Networks | (Foos & de Rooij, 2017) | (Foos & de Rooij, 2016) |
| Do Politicians Racially Discriminate Against Constituents? A Field Experiment on State Legislators | (Butler & Broockman, 2011a) | (Butler & Broockman, 2011b) |
| Community and the Crime Decline: The Causal Effect of Local Nonprofits on Violent Crime | (Sharkey et al., 2017) | (Sharkey, 2018) |
| Interpreting performance information: Motivated reasoning or unbiased comprehension? A replication and extension | (Brogaard et al., 2024) | (Brogaard et al., 2023) |
| The Minimum Legal Drinking Age and Crime | (Carpenter & Dobkin, 2015) | (Carpenter & Dobkin, 2014) |
| Countering misinformation via WhatsApp: Preliminary evidence from the COVID-19 pandemic in Zimbabwe | (Bowles et al., 2020a) | (Bowles et al., 2020b) |
| Bayesian Nonparametric Modeling for Causal Inference | (Hill, 2011) | (Liu et al., 2024) |
| Estimating individual treatment effect: generalization bounds and algorithms | (Shalit et al., 2017) | (Liu et al., 2024) |

Table 13. Publications and the corresponding sources from which we obtained the data.

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